



 POLITECNICO DI MILANO



# Machine Learning

Andrea Mor

[andrea.mor@polimi.it](mailto:andrea.mor@polimi.it)



## How would you rate your Machine Learning knowledge?

[pollev.com/andreamor103](https://pollev.com/andreamor103)





## Artificial Intelligence

Is the study of "intelligent agents", often represented by machines, which mimic cognitive human functions (perceive the environment and take actions) to obtain a specific goal.

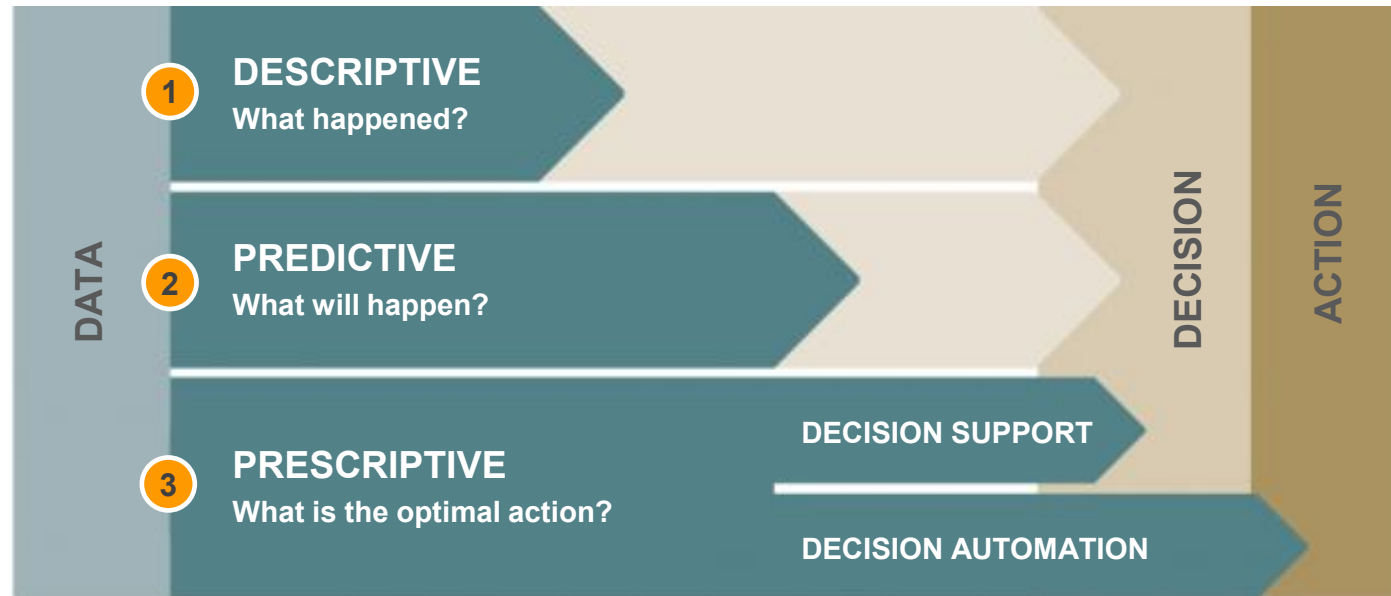
## Machine Learning

Subset of AI focused on algorithms that enable computers to learn from data and improve over time without explicit programming.

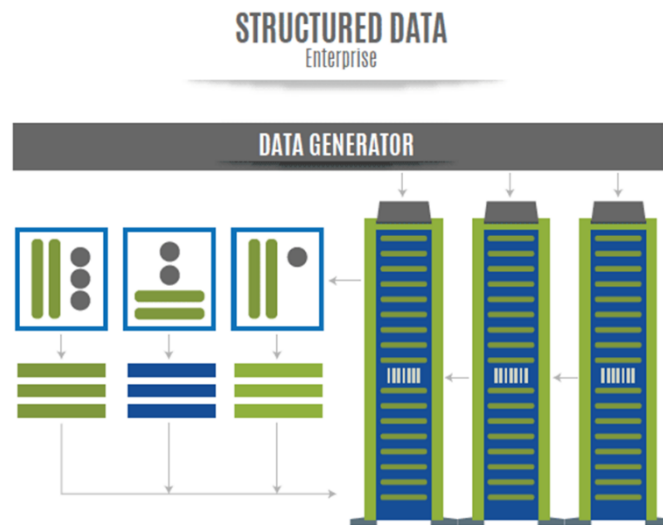


# From descriptive to prescriptive analytics

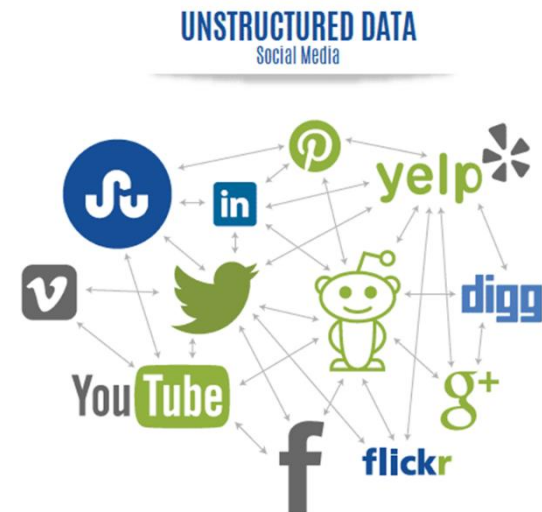
Scheme adapted from the original work by Gartner



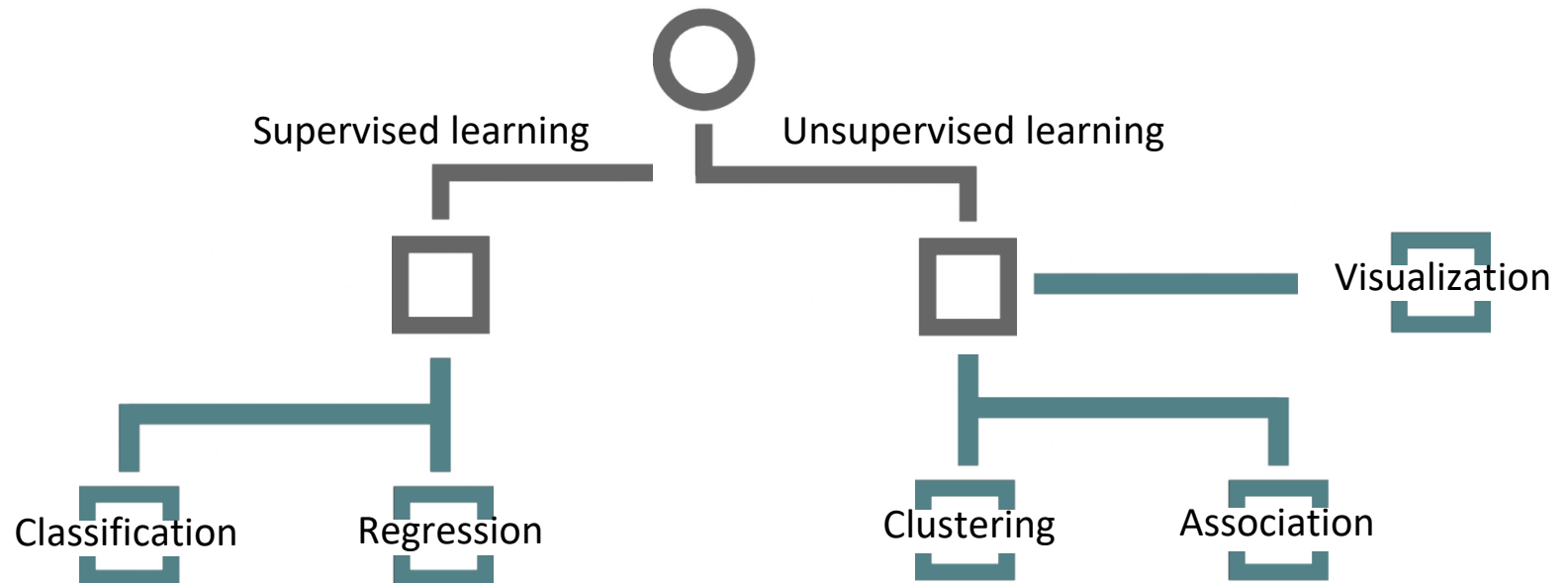
- 1 Analyze historical data to look for the reasons behind a given phenomenon.
- 2 Use algorithms to discover patterns in historical data and predict the future outcome of an event. Patterns should be novel, potentially useful, understandable.
- 3 Suggest the best actions to take advantage of predictions.

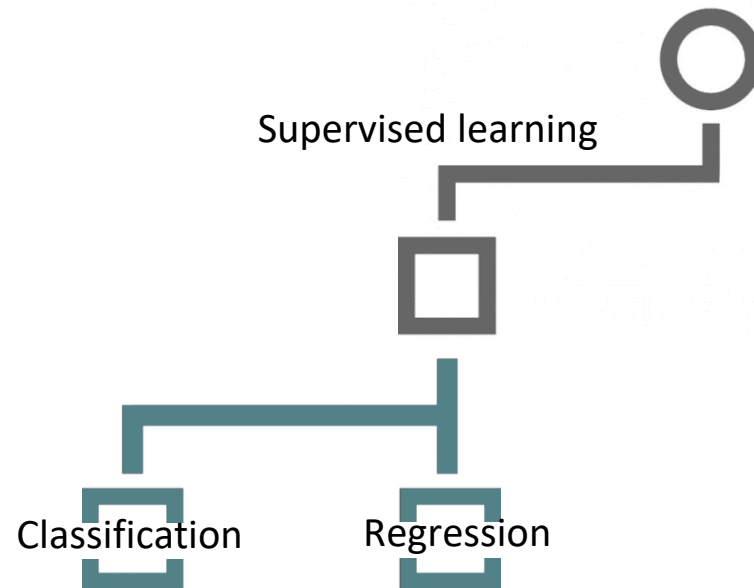


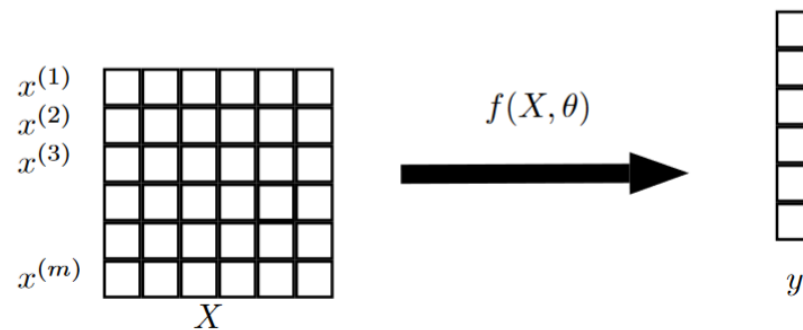
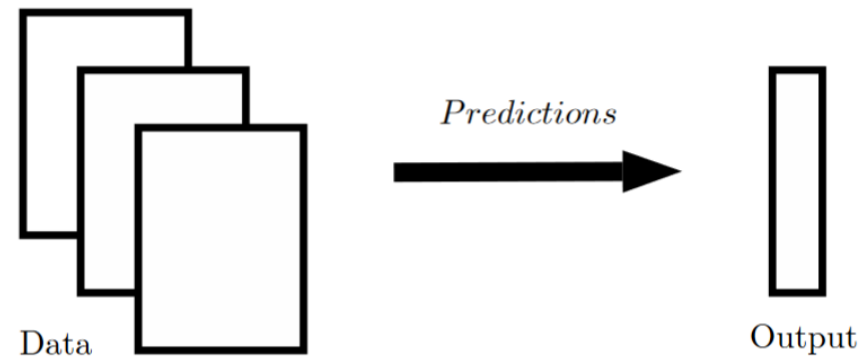
- ✓ Traffic data
- ✓ Commercial transactions
- ✓ Sensor data (GPS, medical devices, smart meters, etc.)
- ✓ Web server logs
- ✓ Financial data
- ✓ ...



- ✓ Social media posts
- ✓ Digital images and videos
- ✓ Audio files
- ✓ ...







Supervised learning:

Given a set  $D$  of data points  $D = \{\mathbf{X}_i, y_i\}$  ( $i \in M$ ), infer a function  $y_i = f(\mathbf{X}_i)$  mapping the inputs to the output.

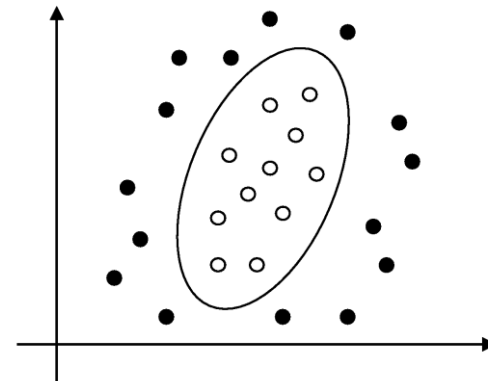




## CLASSIFICATION

**Divide a dataset of examples into classes according to their characteristics with the aim of predicting the class of novel future examples.**

Example: Divide patients into two classes based on the presence/absence of a disease and identify the individuals who are most at risk for prevention purposes.

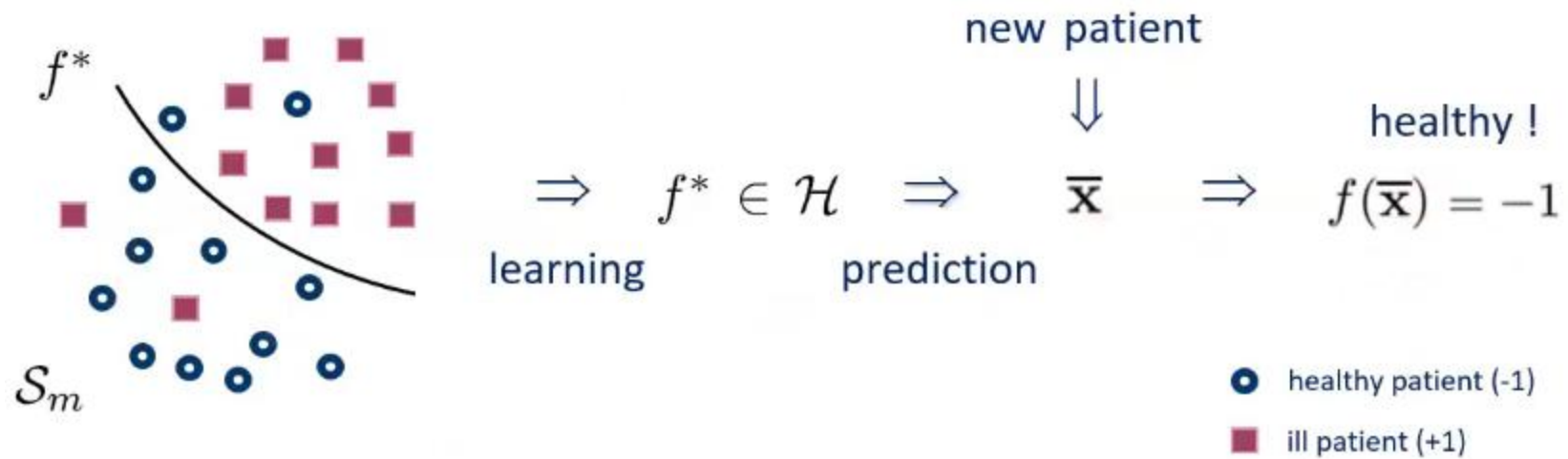




$\mathcal{S}_m = \{(\mathbf{x}_i, y_i), i \in \mathcal{M}\}$  : training set, where  $\mathbf{x}_i \in \mathbb{R}^n$  and  $y_i \in \mathcal{D}$

$\mathcal{H}$  denotes a set of functions  $f(\mathbf{x}) : \mathbb{R}^n \mapsto \mathcal{D}$

**Classification problem:** define a hypotheses space  $\mathcal{H}$  and a function  $f^* \in \mathcal{H}$  which optimally describes the relationship between  $\mathbf{x}_i$  and  $y_i$





## REGRESSION

**Predict the value of a numeric variable for each example in a dataset based on the values of other characteristics.**

Example: Predict the life expectation of a individual based on different characteristics (socio-demographic factors, life-style, profession, clinical tests,...).

$$Y = f (X_1, X_2, \dots, X_n)$$



The dataset contains information about some failures occurred on a machine while processing some products.

| Product ID | UDI | Type | Air temperature [K] | Process temperature [K] | Rotational speed [rpm] | Torque [Nm] | Tool wear [min] | Failure Type       | Failure |
|------------|-----|------|---------------------|-------------------------|------------------------|-------------|-----------------|--------------------|---------|
| L47183     | 4   | L    | 298.2               | 308.6                   | 1433                   | 39.5        | 7               | No Failure         | 0       |
| M14865     | 6   | M    | 298.1               | 308.6                   | 1425                   | 41.9        | 11              | No Failure         | 0       |
| L47186     | 7   | L    | 298.1               | 308.6                   | 1558                   | 42.4        | 14              | No Failure         | 0       |
| L47213     | 34  | L    | 298.9               | 309.3                   | 1665                   | 32.5        | 93              | No Failure         | 0       |
| L47219     | 40  | L    | 298.8               | 309.1                   | 1350                   | 52.5        | 111             | No Failure         | 0       |
| M14902     | 43  | M    | 298.8               | 309.1                   | 1368                   | 49.1        | 117             | No Failure         | 0       |
| L47230     | 51  | L    | 298.9               | 309.1                   | 2861                   | 4.6         | 143             | Power Failure      | 1       |
| L47234     | 55  | L    | 298.7               | 309.0                   | 1691                   | 30.1        | 154             | No Failure         | 0       |
| M14923     | 64  | M    | 298.9               | 309.0                   | 1812                   | 25.9        | 174             | No Failure         | 0       |
| L47249     | 70  | L    | 298.9               | 309.0                   | 1410                   | 65.7        | 191             | Power Failure      | 1       |
| L47257     | 78  | L    | 298.8               | 308.9                   | 1455                   | 41.3        | 208             | Tool Wear Failure  | 1       |
| M14957     | 98  | M    | 298.9               | 308.9                   | 1750                   | 29.9        | 50              | No Failure         | 0       |
| L47278     | 99  | L    | 298.9               | 308.8                   | 1529                   | 32.7        | 53              | No Failure         | 0       |
| L47296     | 117 | L    | 298.9               | 308.7                   | 1564                   | 34.1        | 99              | No Failure         | 0       |
| H29557     | 144 | H    | 298.6               | 308.4                   | 1407                   | 50.5        | 164             | No Failure         | 0       |
| M15008     | 149 | M    | 298.3               | 308.3                   | 1379                   | 48.0        | 181             | No Failure         | 0       |
| M15018     | 159 | M    | 298.4               | 308.2                   | 1499                   | 40.0        | 211             | No Failure         | 0       |
| L47340     | 161 | L    | 298.4               | 308.2                   | 1282                   | 60.7        | 216             | Overstrain Failure | 1       |

↑  
Quality of the product

↑  
Force that tends to  
cause the rotation  
around an axis

↑  
Time before observing  
the gradual failure of the  
cutting tools



The data set contains the log of bike trips as recorder by the software operating a bike sharing system

| IngPKID | IngFKIDBike | IngFKIDBadge | dtmDateTimePickup   | dtmDateTimeReturn   | IngFKIDStationPickup | intStandPickup | IngFKIDStationReturn | intStandReturn |
|---------|-------------|--------------|---------------------|---------------------|----------------------|----------------|----------------------|----------------|
| 5074    | 213         | 4704         | 2011-08-28 00:04:43 | 2011-08-28 18:12:36 | 22                   | 3              | 20                   | 7              |
| 5106    | 213         | 3298         | 2011-08-28 11:19:04 | 2011-08-28 11:43:07 | 37                   | 4              | 35                   | 13             |
| 5114    | 213         | 5688         | 2011-08-28 12:19:17 | 2011-08-28 12:34:20 | 35                   | 13             | 20                   | 7              |
| 5151    | 213         | 4            | 2011-08-28 18:19:26 | 2011-08-28 18:57:44 | 20                   | 7              | 33                   | 4              |
| 5166    | 213         | 6059         | 2011-08-28 19:49:43 | 2011-08-28 19:56:53 | 33                   | 4              | 37                   | 7              |
| 5177    | 213         | 2534         | 2011-08-28 21:00:57 | 2011-08-28 21:03:42 | 37                   | 7              | 31                   | 4              |
| 5187    | 213         | 5956         | 2011-08-28 22:00:58 | 2011-08-28 22:18:24 | 31                   | 4              | 36                   | 5              |
| 5198    | 213         | 5956         | 2011-08-28 23:17:54 | 2011-08-28 23:22:04 | 36                   | 5              | 31                   | 1              |
| 5203    | 213         | 3861         | 2011-08-29 06:48:53 | 2011-08-29 07:09:19 | 31                   | 1              | 6                    | 8              |
| 5560    | 213         | 3053         | 2011-08-29 17:29:12 | 2011-08-29 17:29:40 | 6                    | 8              | 6                    | 8              |
| 5576    | 225         | 5949         | 2011-08-29 17:44:01 | 2011-08-29 21:01:36 | 25                   | 16             | 25                   | 12             |
| 5583    | 225         | 5949         | 2011-08-29 17:53:58 | 2011-08-29 20:01:03 | 25                   | 27             | 25                   | 12             |
| 5584    | 225         | 5949         | 2011-08-29 17:54:41 | 2011-08-29 17:55:18 | 25                   | 28             | 25                   | 12             |
| 5717    | 213         | 3            | 2011-08-30 07:33:10 | 2011-08-30 07:57:57 | 6                    | 8              | 25                   | 29             |
| 5780    | 225         | 2444         | 2011-08-30 08:43:43 | 2011-08-30 22:52:00 | 25                   | 12             | 30                   | 4              |
| 5807    | 225         | 5778         | 2011-08-30 09:17:12 | 2011-08-30 09:21:32 | 20                   | 8              | 19                   | 1              |
| 5814    | 213         | 5516         | 2011-08-30 09:27:58 | 2011-08-30 09:33:43 | 22                   | 13             | 38                   | 8              |
| 5836    | 213         | 5697         | 2011-08-30 10:06:36 | 2011-08-30 10:20:55 | 38                   | 8              | 6                    | 7              |
| 5908    | 213         | 5474         | 2011-08-30 11:59:05 | 2011-08-30 12:19:12 | 6                    | 7              | 38                   | 1              |
| 5916    | 225         | 3644         | 2011-08-30 12:06:28 | 2011-08-30 20:06:13 | 19                   | 1              | 37                   | 9              |
| 5948    | 225         | 3644         | 2011-08-30 12:52:35 | 2011-08-30 18:25:25 | 19                   | 1              | 37                   | 9              |



What is the target in this case?

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Can you think of some more applications for these methodologies?

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## Retail

Find segments of customers who are likely to respond to a marketing campaign.

Forecast the demand for a given product.

Find which products are often bought together.

## Banking

Find segments of individuals at high risk of insolvency (granting a loan).

Identify anomalous transactions.

## Insurance

Identify fraudulent behaviours.

Find segments of customers who are likely to churn.

## Healthcare

Find groups of patients who are more likely to be re-hospitalized.

Find genes which are highly informative against the presence of a given pathology.

## Manufacturing

Build models for early failure detection or quality control.

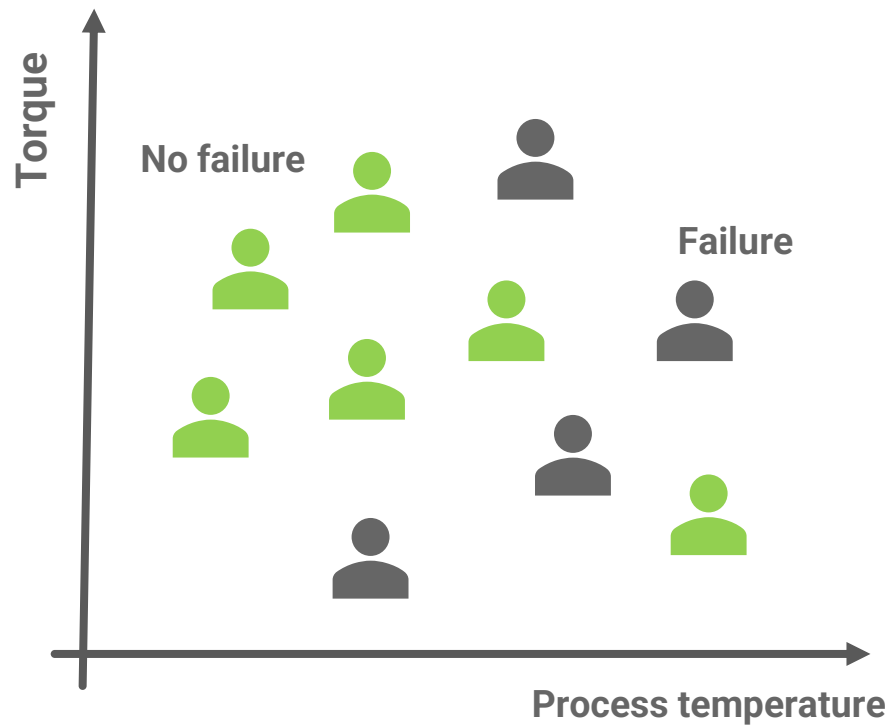
## ...cross-domain analysis

Image and video recognition, text classification (sentiment analysis), signal recognition, etc...



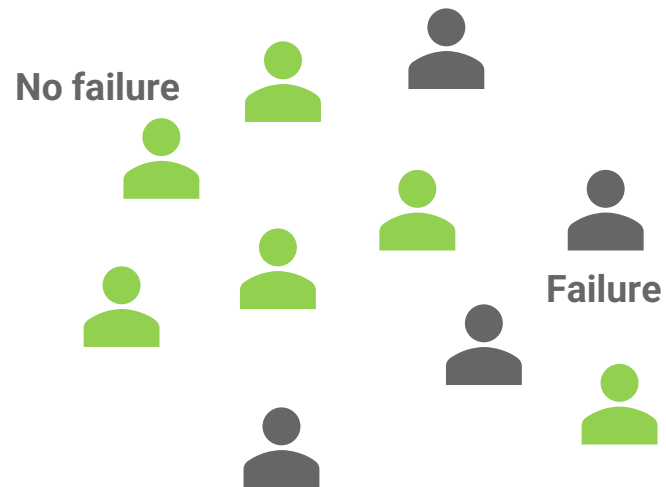


Examples are points in the n-dimensional feature space.



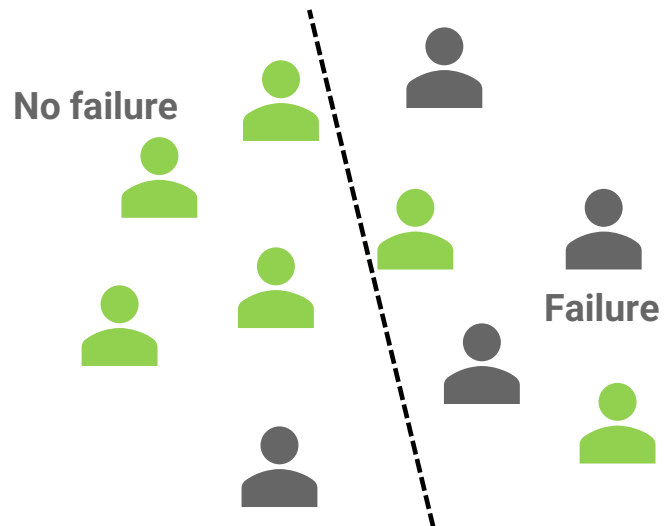


# Classification: how to build the model



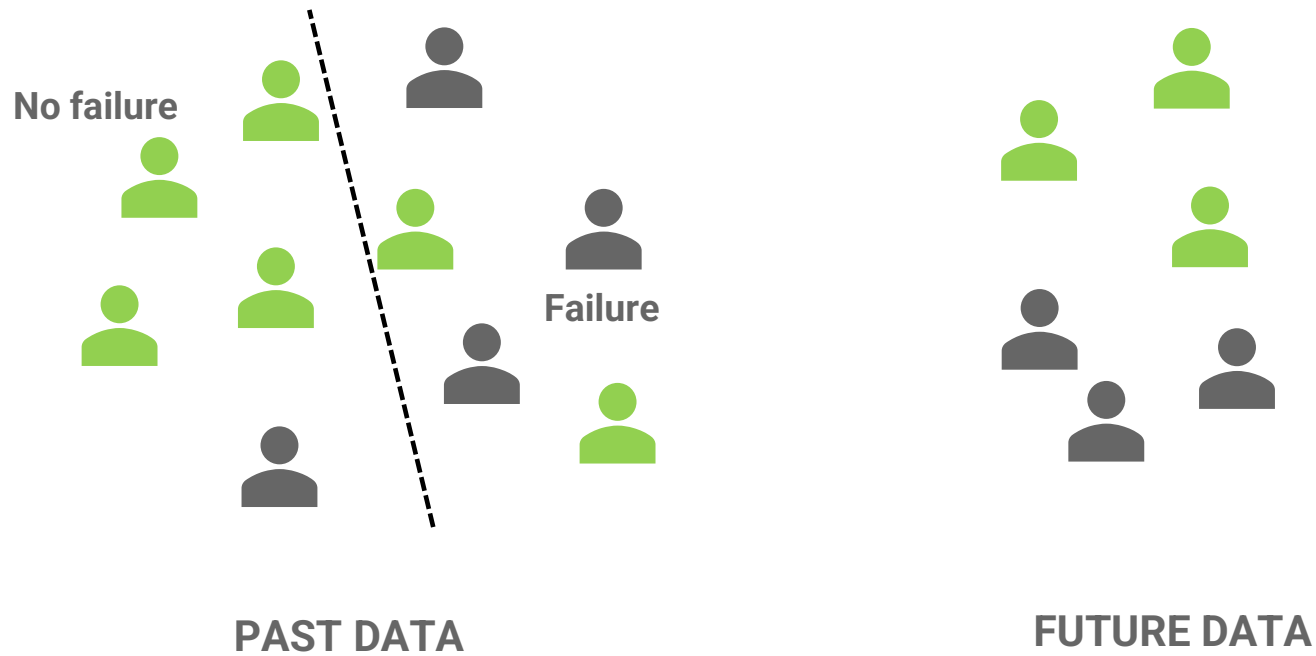


# Classification: how to build the model





# Classification: how to build the model



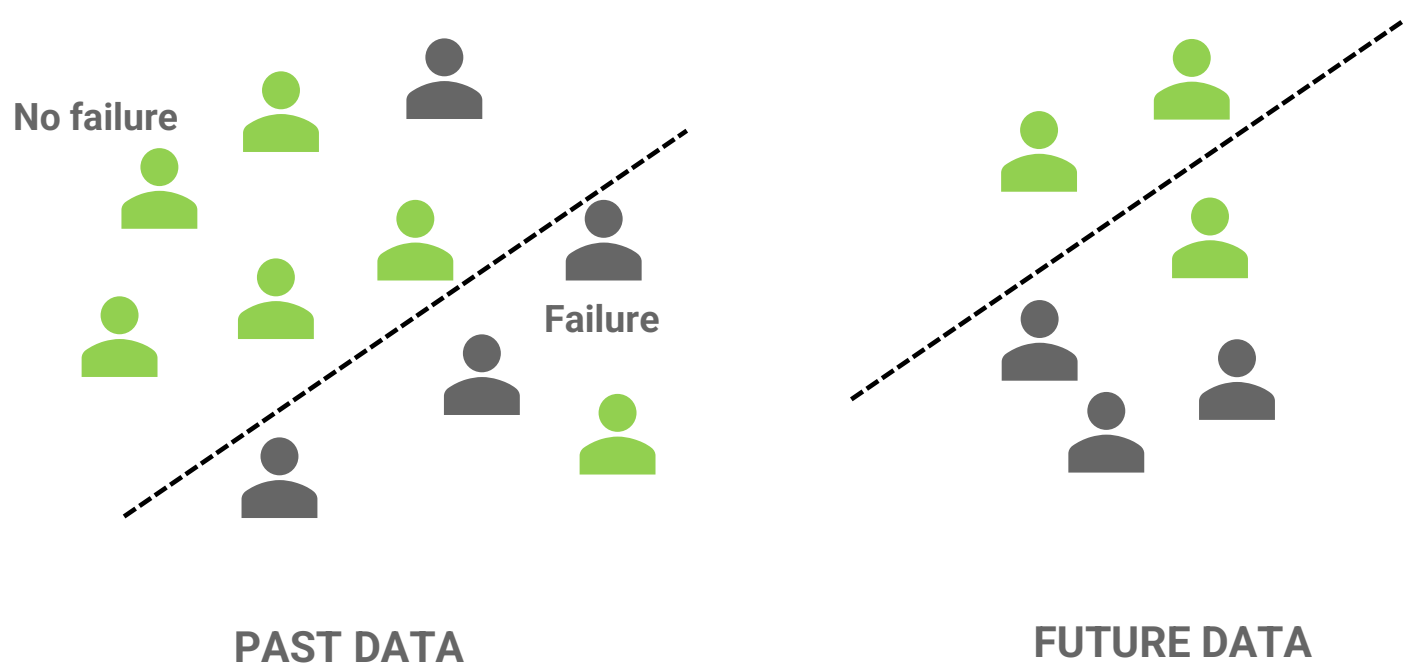


# Classification: how to build the model



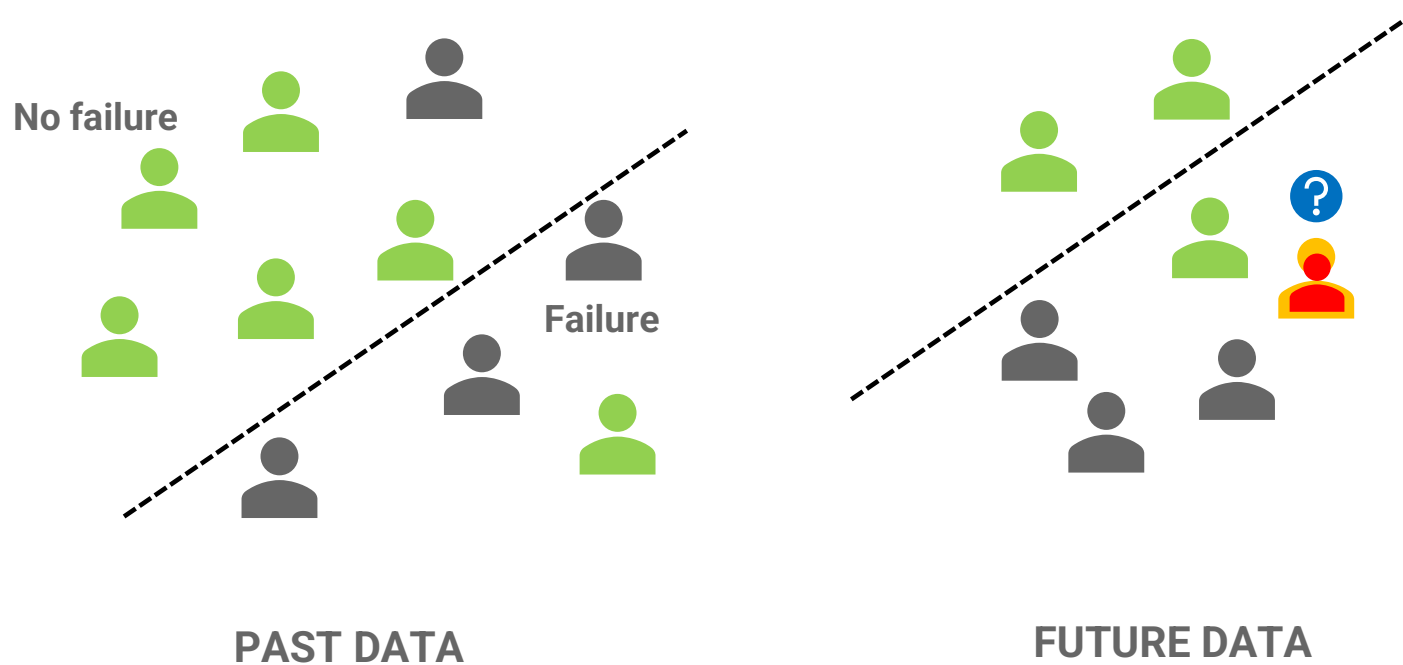


Different models could be generated.



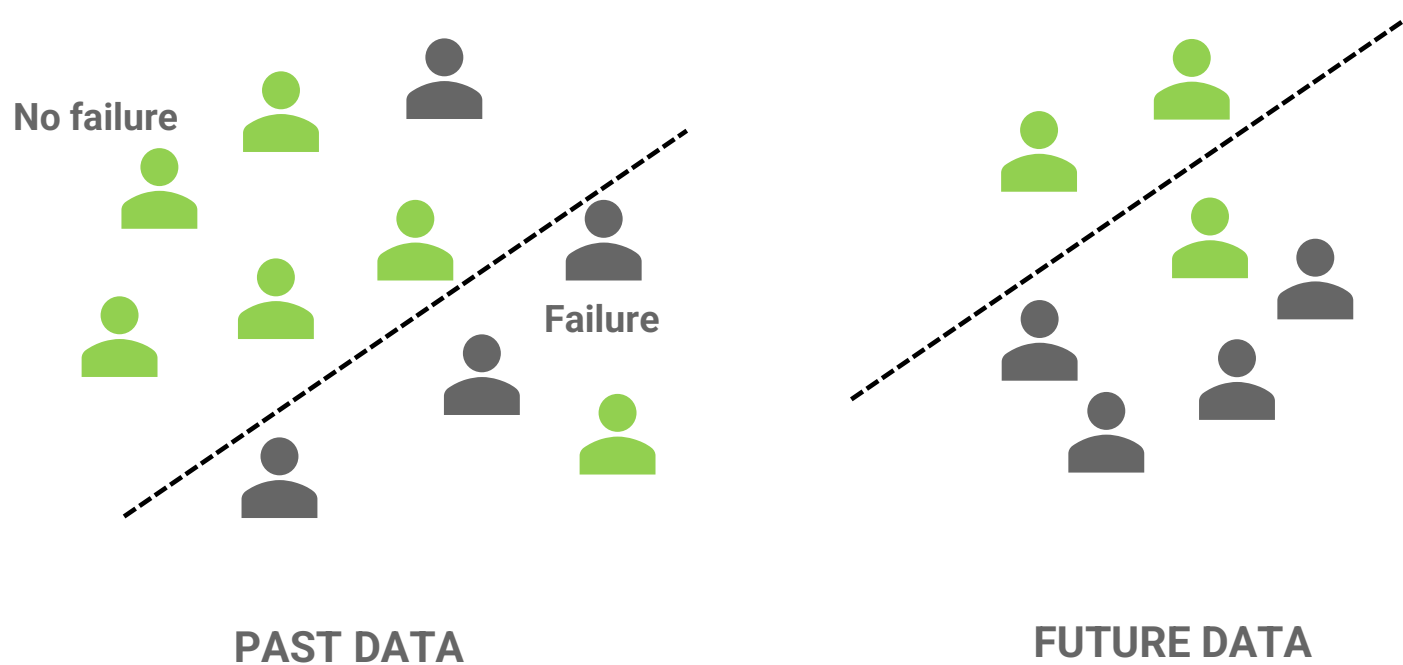


# Classification: how to build the model





# Classification: how to build the model







Most algorithms generate a score which is a proxy of the probability of belonging to a given class.

|   | true_class | pred_class | no         | yes        |
|---|------------|------------|------------|------------|
| 1 | no         | no         | 1.00000000 | 0.00000000 |
| 2 | no         | no         | 1.00000000 | 0.00000000 |
| 3 | yes        | yes        | 0.13333333 | 0.86666667 |
| 4 | no         | no         | 0.84615385 | 0.15384615 |
| 5 | no         | no         | 1.00000000 | 0.00000000 |
| 6 | yes        | yes        | 0.38461538 | 0.61538462 |
| 7 | no         | no         | 0.92307692 | 0.07692308 |
| 8 | yes        | yes        | 0.00000000 | 1.00000000 |
| 9 | no         | no         | 0.84615385 | 0.15384615 |

The cutoff value can be properly tuned to effectively classify unbalanced datasets.

Fit a classification model, generate the probability of a test example being in the “yes” class and then play with the cutoff point at which you draw the line between “yes” and “no”.



## How would you assess the quality of a model?

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How can we evaluate the quality of a classification model?

Let's introduce the following **LOSS FUNCTION**

$$L(y_i, f(\mathbf{x}_i)) = \begin{cases} 0 & \text{if } y_i = f(\mathbf{x}_i) \\ 1 & \text{otherwise} \end{cases}$$

and compute the accuracy...

$$\text{acc}_A(\mathcal{S}) = 1 - \frac{1}{m} \sum_{i=1}^m L(y_i, f(\mathbf{x}_i))$$

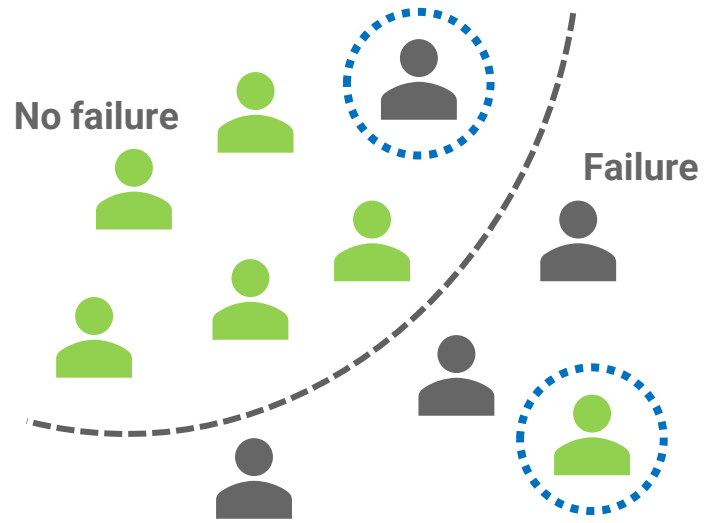
↑                      ↑  
Empirical error

► **Accuracy** =  $\frac{TP+TN}{TP+FP+TN+FN}$

|              |   | Prediction outcome |                |
|--------------|---|--------------------|----------------|
|              |   | 0                  | 1              |
| Actual value | 0 | True Negative      | False Positive |
|              | 1 | False Negative     | True Positive  |

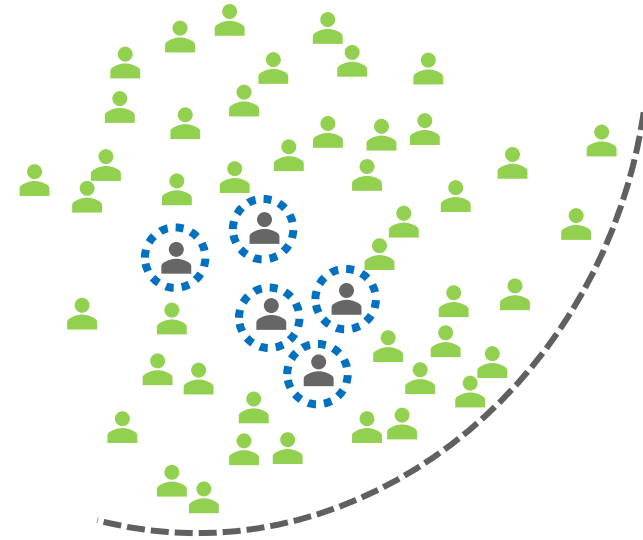


# Is accuracy enough?



Accuracy = 80%

Error = 100 – Accuracy = 20%



Accuracy = 90%

Error = 10%

|              |   | Prediction outcome |                |
|--------------|---|--------------------|----------------|
|              |   | 0                  | 1              |
| Actual value | 0 | True Negative      | False Positive |
|              | 1 | False Negative     | True Positive  |

- ▶ **Accuracy** =  $\frac{TP+TN}{TP+FP+TN+FN}$   
**When:** balanced problems, equal costs of FN and FP.  
**When NOT:** imbalanced problems.
- ▶ **Recall** (True Positive rate) =  $\frac{TP}{FN+TP}$   
“proportion of true positives among actual positive”  
**When:** you want to be sure to catch all the 1s, at the cost of having false alerts.
- ▶ **Precision** (Positive Predictive value) =  $\frac{TP}{TP+FP}$   
“proportion of true positives among positive predictions”  
**When:** you want to be sure that all that you predict as 1, are indeed 1, e.g., false alerts are costly.
- ▶ Precision and Recall are typically evaluated together.

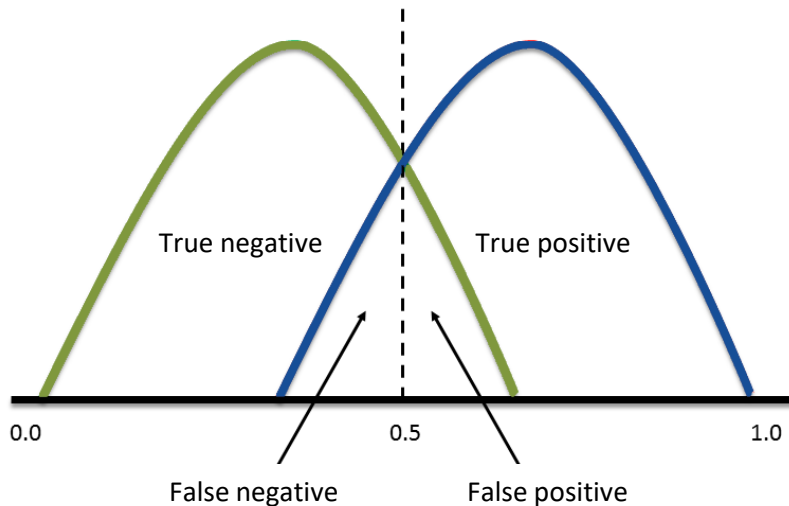
|              |   | Prediction outcome |                |
|--------------|---|--------------------|----------------|
|              |   | 0                  | 1              |
| Actual value | 0 | True Negative      | False Positive |
|              | 1 | False Negative     | True Positive  |

- ▶ **Geom. mean** =  $\sqrt{\text{Precision} \times \text{Recall}}$
- ▶ **F-score** =  $\frac{(1+\beta^2)}{\beta^2} \frac{1}{\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}} = (1 + \beta^2) \frac{\text{Precision} \cdot \text{Recall}}{(\beta^2 \cdot \text{Precision}) \cdot \text{Recall}}$
- ▶ **F1**: harmonic mean of Prec. and Recall  
$$= \frac{2TP}{2TP + FP + FN}$$

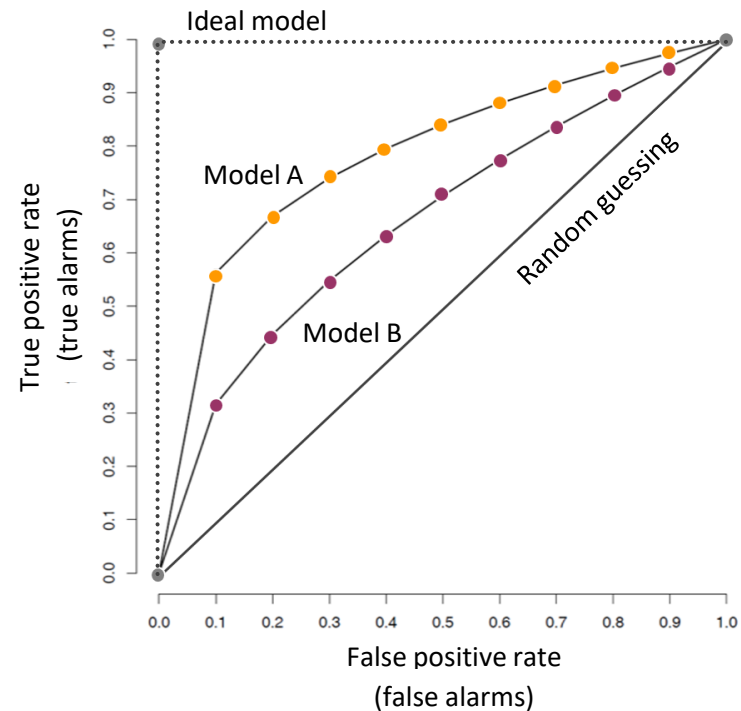
Frequently the to-go metric.
- ▶ **F2**: 2x emphasis on Recall  
**When:** False Negatives are more costly.



Any probability threshold is going to produce true positives, false positives, true negatives and false negatives.



ROC analysis detects possibly optimal models and discard suboptimal ones independently from the class distribution.



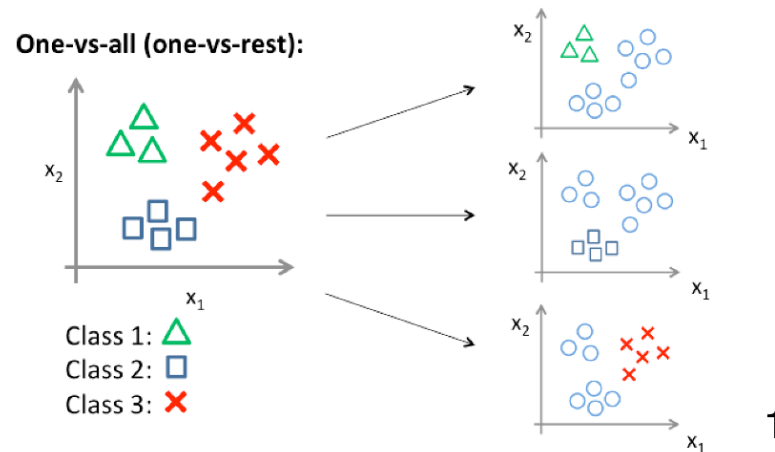
A ROC curve can be used to select a threshold for a classifier according to the specific needs (i.e., maximize TP rate or minimize FP rate).





# Does classification need to be binary?

1. **One-vs-Rest** We perform  $|H|$  different binary classifications: one for every class.

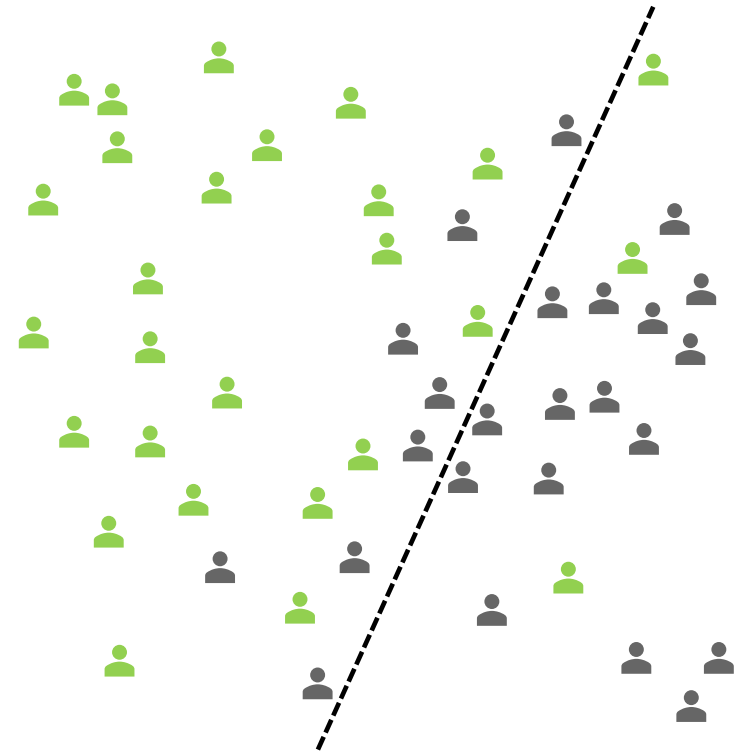


We decide based on a majority vote.

2. **One-vs-One** We perform  $|H|(|H| - 1)/2$  binary classifications: one for every pair of classes. We decide based on a majority vote.

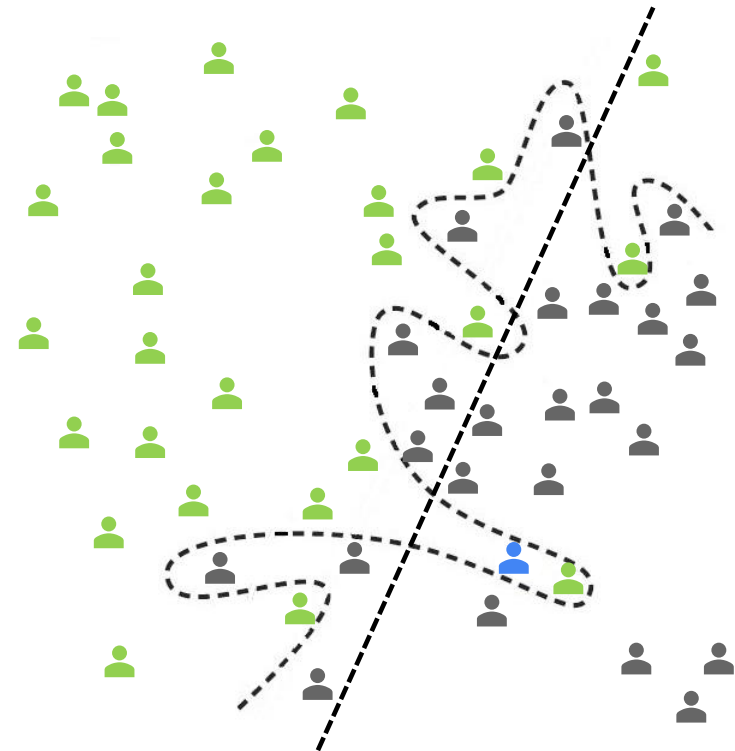


Overfitting occurs when a model learns noise or random fluctuations in the training data so that its performance on new data (generalization ability) is negatively affected. It is often a result of an excessively complicated model.



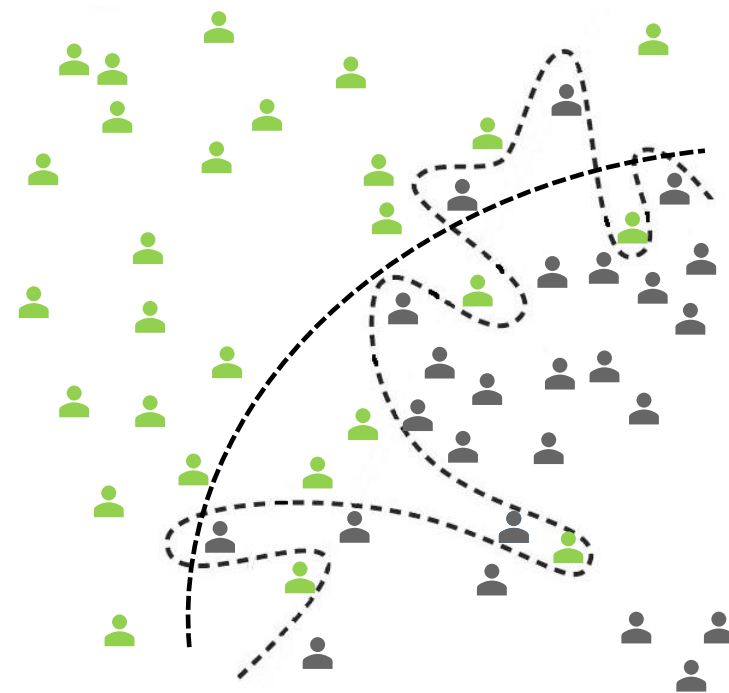


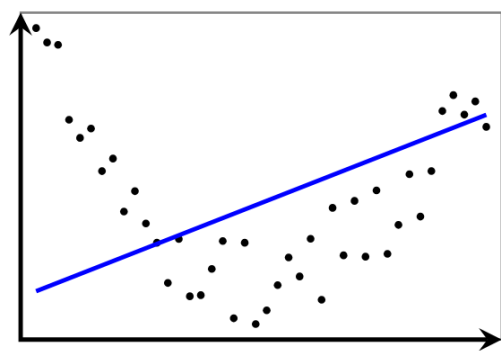
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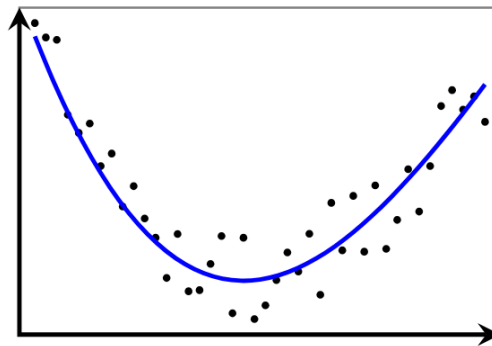


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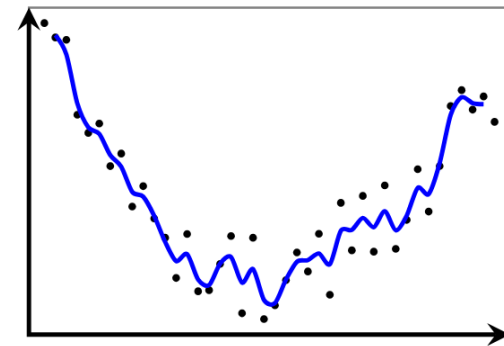




Underfitting



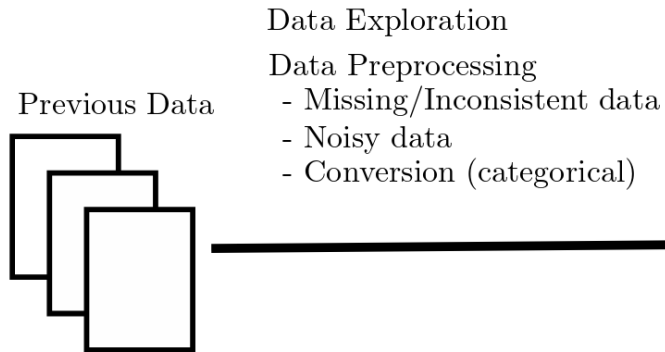
Balance



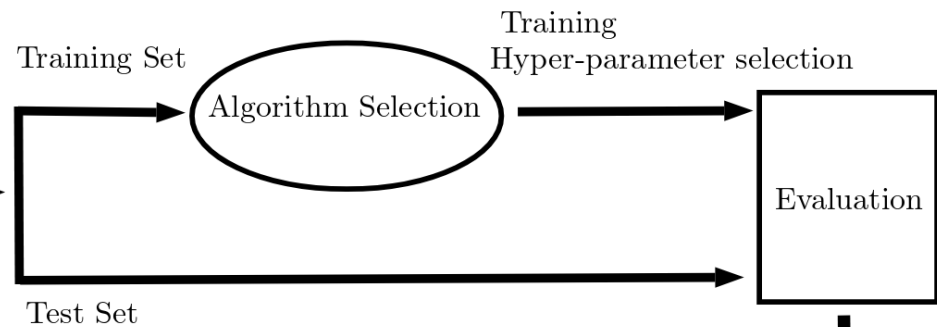
Overfitting



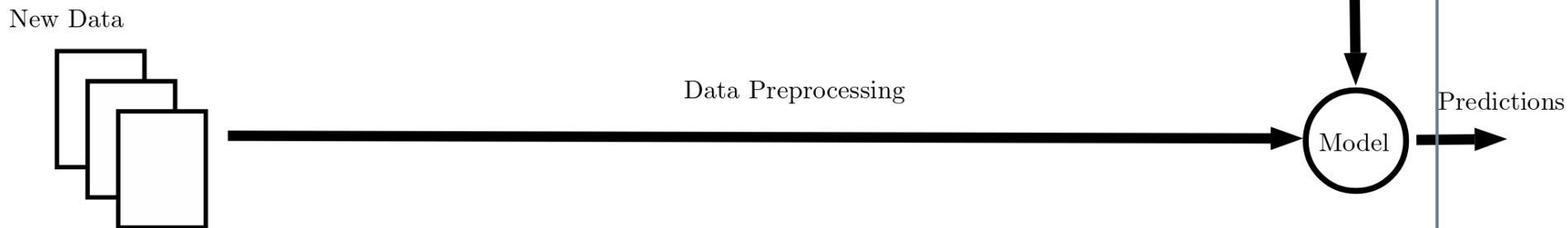
## 1. Data Exploration/Analysis



## 2. Model Creation



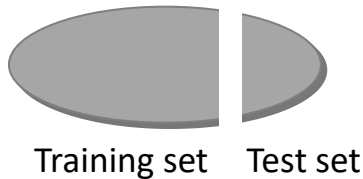
## 3. Predictions



Accuracy can be estimated by means of four different evaluation schemes:

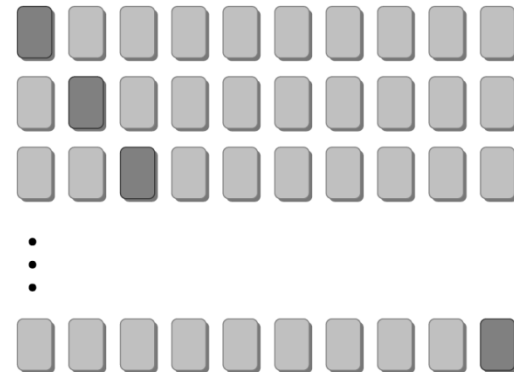
## ① HOLDOUT METHOD

The available examples are divided in two disjoint subsets used for different purposes.



## ③ K-FOLD CROSS-VALIDATION

The available examples are divided in two K disjoint subsets.



## ② REPEATED RANDOM SAMPLING

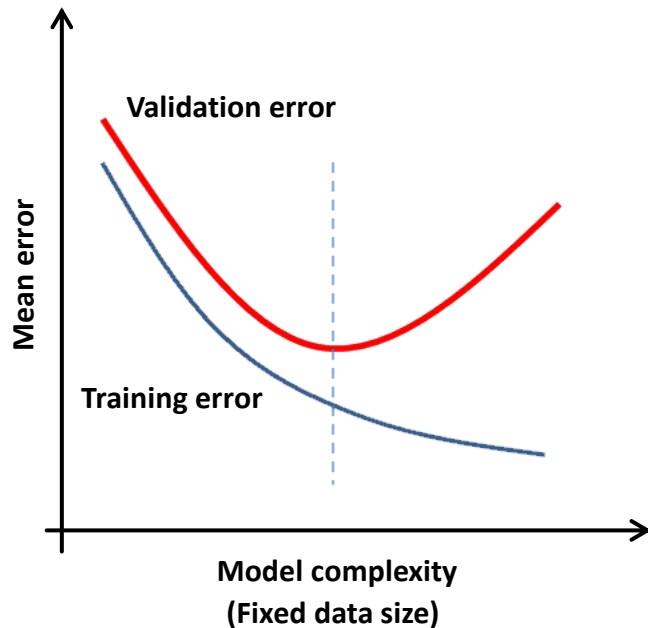
Holdout method repeated several times.

## ④ LEAVE-ONE-OUT

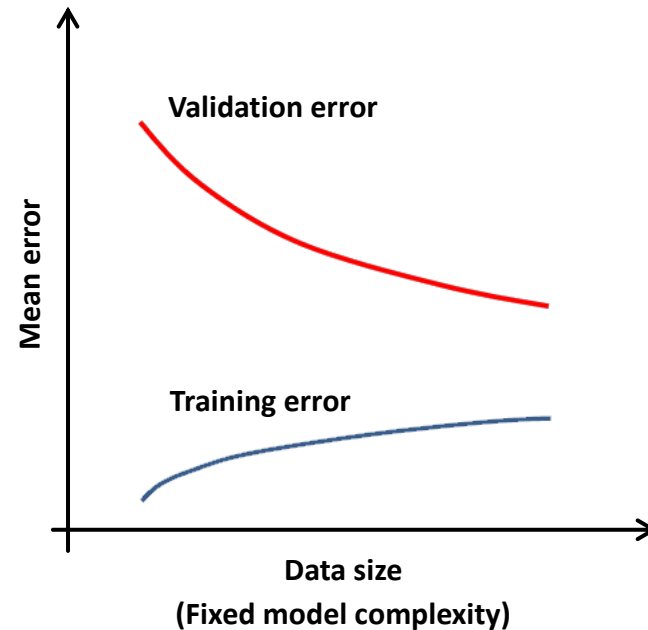
K-fold cross-validation with  $K = m$ .



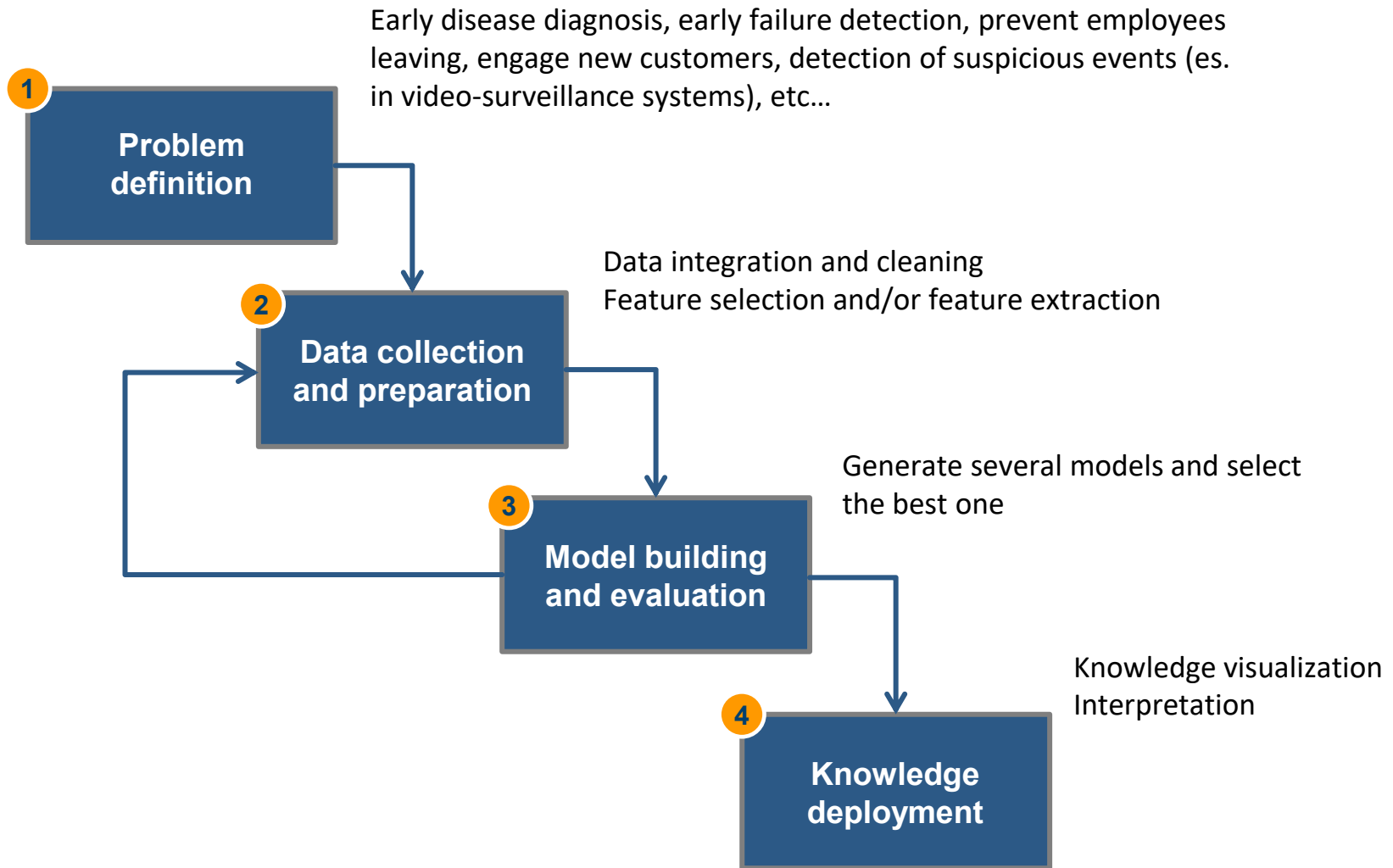
When the size of training data is fixed and the only thing we can choose is the model complexity, then we should choose the model complexity at the point where the cross-validation error is minimal.

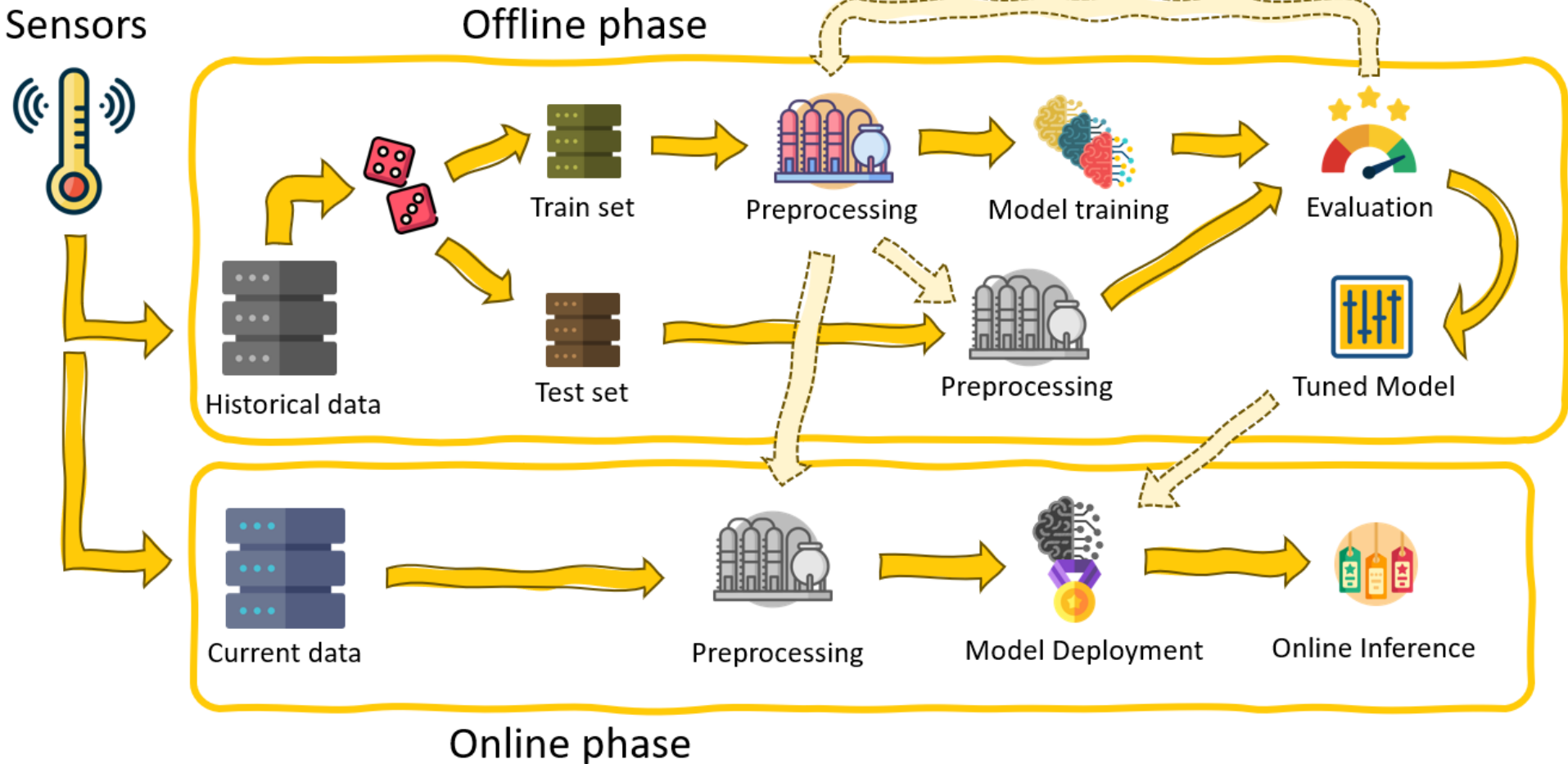


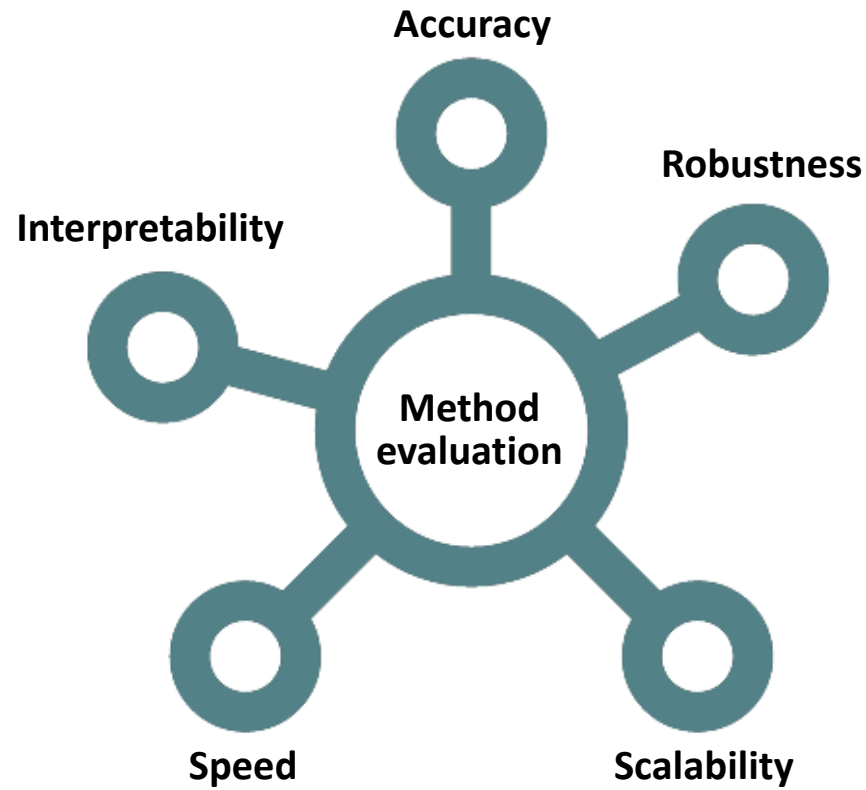
When the underlying data model is in fact complex we cannot reduce the complexity (over-simplified model) but we can collect more training data such that overfitting is less likely to happen (or using bagging).







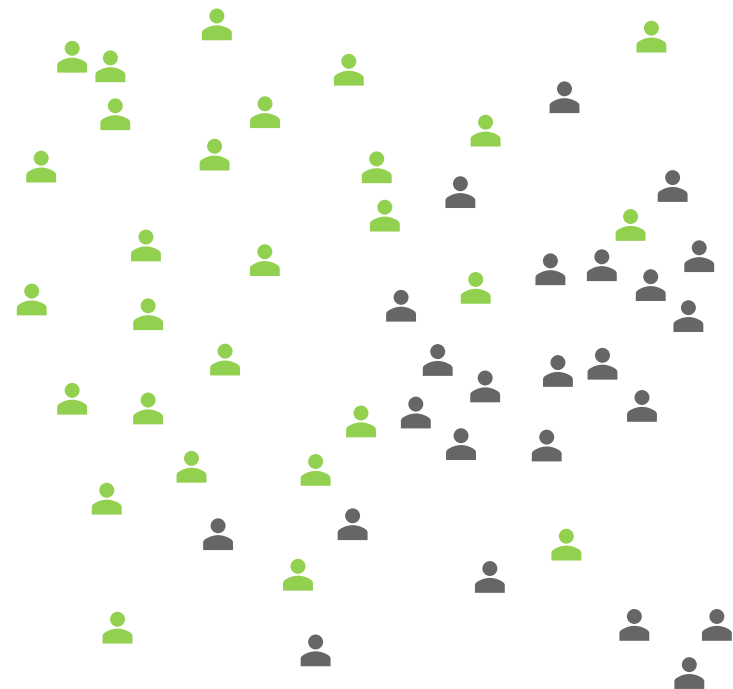






## How would you build a classification model?

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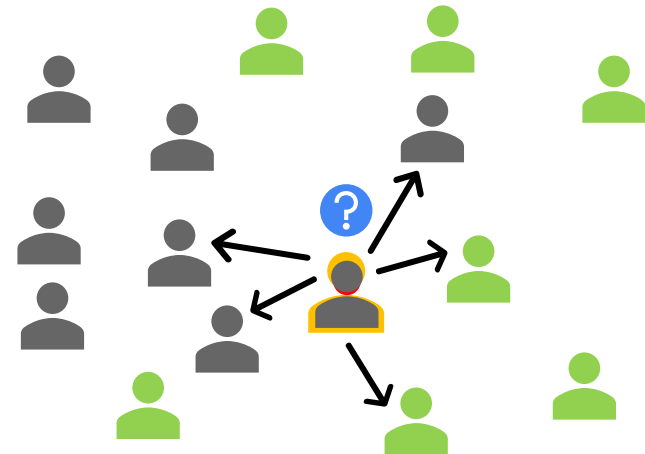


# Nearest neighbors classifier (K-NN)

The **K-NEAREST NEIGHBORS** classifier is a *non-parametric\** algorithm used for classification (and regression). It is in the family of *instance-based learning* methods.

A set of training examples must be given. The class of a new example is computed based on the class value of its K nearest neighbors (*majority voting*).

The choice of K affects the accuracy of the model. Rule of thumb:  $k = \text{radq}(m)$ .

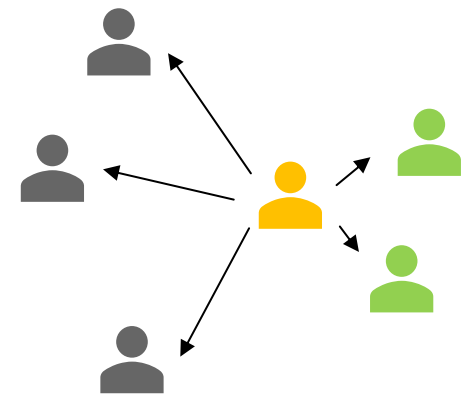


\*No assumptions on the distribution from which the training examples are drawn.



Some comments:

- ① Conceptually simple
- ② It works well with few examples and/or in low dimensions for complex decision surfaces
- ③ Classification might be slow (computationally intensive)
- ④ It suffers from the *curse of dimensionality*\*.  
In high dimensions the ratio between the nearest and the farthest point approaches 1 → points become uniformly distant from each other and any kind of distinction becomes meaningless.
- ⑤ Implicit assumption: the class of a new example is close to the classes of its neighbors

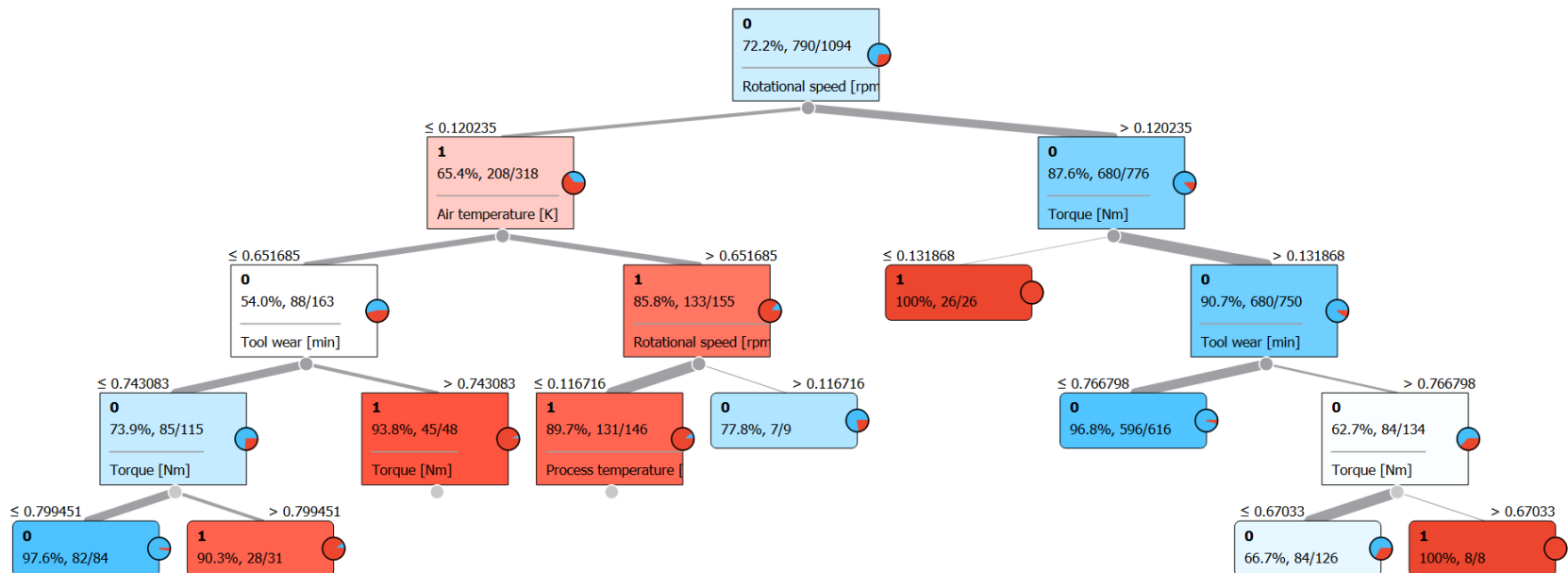


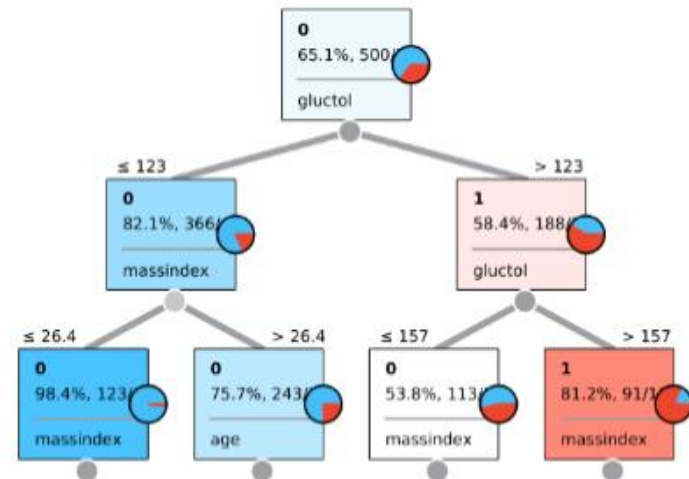
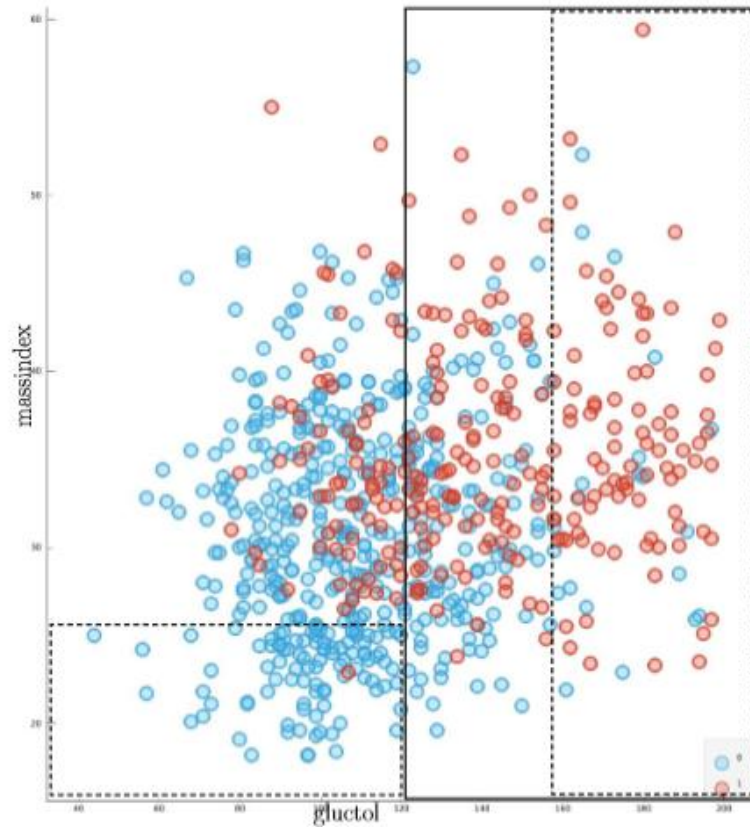
\*With an increase in the dimensions the available data become sparse. The amount of data should grow (exponentially) with dimensions.



CLASSIFICATION TREES resort to an iterative heuristic procedure based on a *divide-and-conquer* scheme

- 1 All examples are included in the root node
- 2 The examples in each node are divided according to a test based on the value of one or more variables
- 3 A node becomes a leaf if it satisfies one of the stopping conditions
- 4 Each leaf is labeled according majority voting



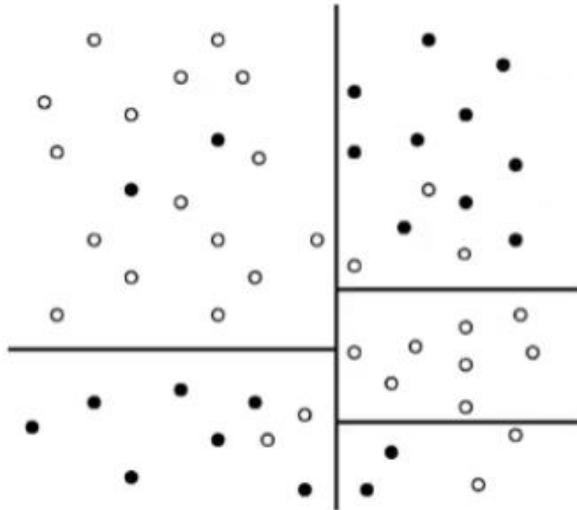


Tree

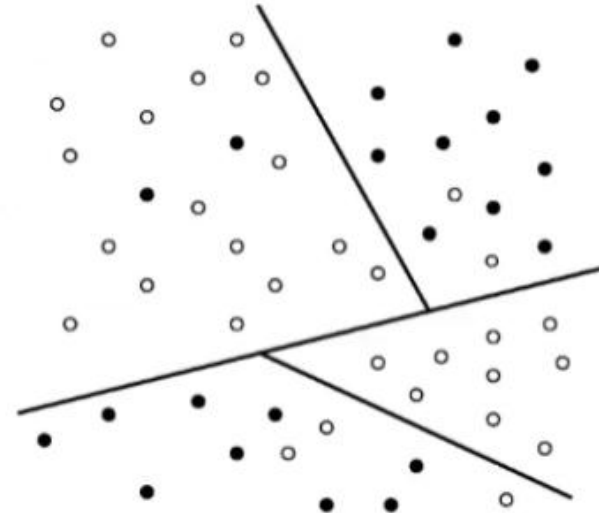
## types

- ▶ Binary tree (zero/two descendants)
- ▶ General trees
- ▶ Uni-variate tree ( $X_j < b$ )
- ▶ Multi-variate tree ( $\sum_{j=1}^n w_j x_j < b$ )

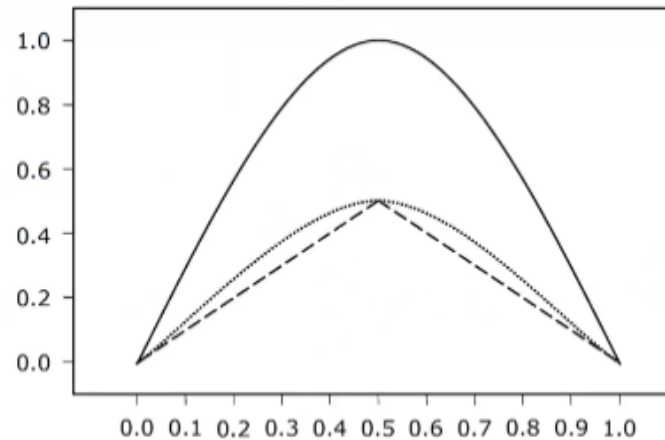
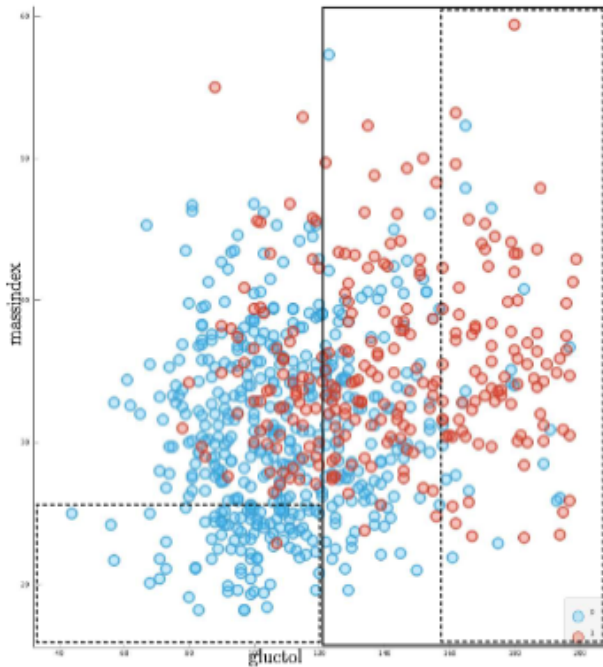




classification by an  
axis parallel tree



classification by an  
oblique tree



- ▶ Entropy index:
- ▶ Gini index:
- ▶ Miss-classification index:

**Split criteria**

$$\begin{aligned} & -\sum_{h=1}^H f_h \log_2 f_h \\ & 1 - \sum_{h=1}^H f_h^2 \\ & 1 - \max_h f_h \end{aligned}$$

$$\text{Information gain} = H_p - \sum_{c \in C} \frac{|N_c| H_c}{|N|}$$



Some comments:

- ① Classification rules are easy to interpret and explain
- ② Implicitly perform variable feature selection (variables used at the top are in general the most important ones)
- ③ Are flexible (able to handle both numerical and categorical data)
- ④ Usually benefit from data discretization
- ⑤ They tend to overfit the training data unless some mechanisms are employed to avoid over-complex trees (*pruning*).
  - ⑤a Pre-pruning → Stopping criteria (number of examples, purity of the node)
  - ⑤b Post-pruning → A node is converted into a leaf if the pruned tree makes fewer error on the pruning set than the original tree.



**BAYESIAN CLASSIFIERS** are aimed at determining the probability that a given example belongs to a class.

► Bayes Theorem

$$P(y|\mathbf{x}) = \frac{P(\mathbf{x}|y)P(y)}{P(\mathbf{x})} = \frac{P(\mathbf{x}|y) P(y)}{\sum_{l=1}^H P(\mathbf{x}|y) P(y)}$$

► Maximum a posteriori hypothesis

$$y_{MAP} = \arg \max_{y \in \mathcal{H}} P(y|\mathbf{x}) = \arg \max \frac{P(\mathbf{x}|y)P(y)}{P(\mathbf{x})}$$



- Categorical/discrete attributes

$$P(x_j|y) = P(x_j = r_{jk}|y = v_h)$$

→ empirical frequency of the observed value on the class  $v_h$

- Numerical attribute

$$P(x_j|y) \sim N(\mu_{jh}, \sigma_{jh})$$

→ assuming Gaussian density with empirical parameters



Some comments:

- ① The conditional independence assumption means that interactions between variables can be ignored and are not modeled:
  - ①a It employs a very simple hypothesis function (sometimes too simple)
  - ②a It ignores some of the information in the training data (no overfitting)
  - ③a Requires less training data. Indeed, it has been shown to perform very well with very small training datasets compared to most of the other classifiers

Bayes Belief Network attempt to overcome such limitations.



The model postulates that

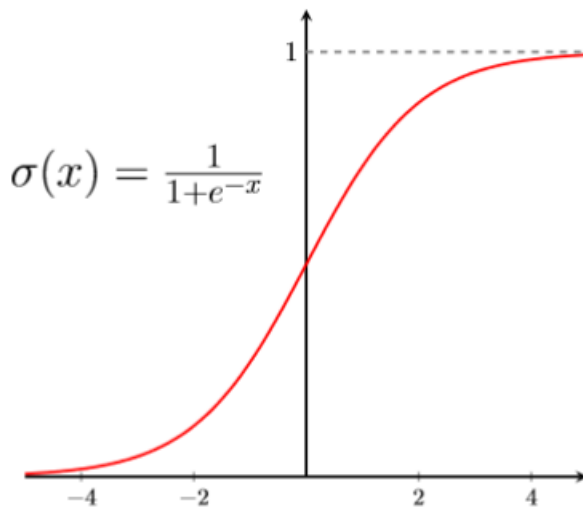
$$\log \left[ \frac{P(y = 1|x)}{P(y = 0|x)} \right] = w^T x$$

then if  $p = P(y = 1|x)$ :

$$\frac{p}{1-p} = e^{w^T x}$$

$$\Rightarrow p = P(y = 1|x) = \frac{e^{w^T x}}{1 + e^{w^T x}} = \frac{1}{1 + e^{-w^T x}},$$

$$P(y = 0|x) = 1 - p = \frac{1}{1 + e^{w^T x}} = \frac{e^{-w^T x}}{1 + e^{-w^T x}}$$





Therefore if we maximize the likelihood

$$L(w) := P(Y_1 = y_1, \dots, Y_m = y_m | w, x_1, \dots, x_m) = \prod_{i=1}^n P(Y_i = y_i | w, x_1, \dots, x_m).$$

Assuming independence:

$$L(w) := \prod_{i|y_i=1} p(x_i) \cdot \prod_{i|y_i=0} (1 - p(x_i))$$

$$L(w) := \prod_{i=1}^n p(x_i)^{y_i} (1 - p(x_i))^{1-y_i}$$

is equivalent to maximize the log likelihood

$$\begin{aligned} l(w) &= \sum_{i=1}^n (y_i \log(p(x_i)) + (1 - y_i) \log(1 - p(x_i))) \\ &= \sum_{i=1}^n (y_i \log\left(\frac{p(x_i)}{1 - p(x_i)}\right) + \log(1 - p(x_i))) \end{aligned}$$

Given the quantities defined above

$$l(w) = \sum_{i=1}^n (y_i w^T x_i - \log(1 + e^{-w^T x_i}))$$

Therefore we aim to resolve the following optimization problem

$$\max_w l(w)$$

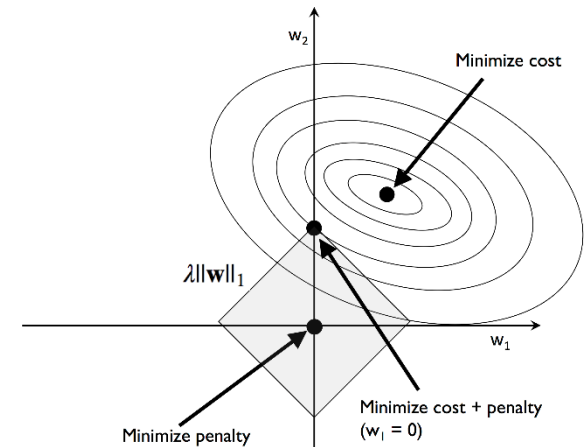
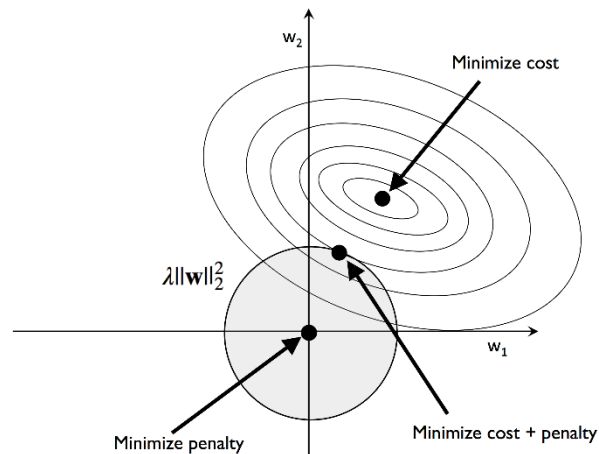
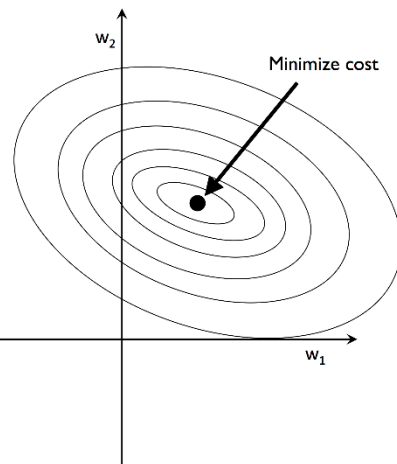




## Regularization

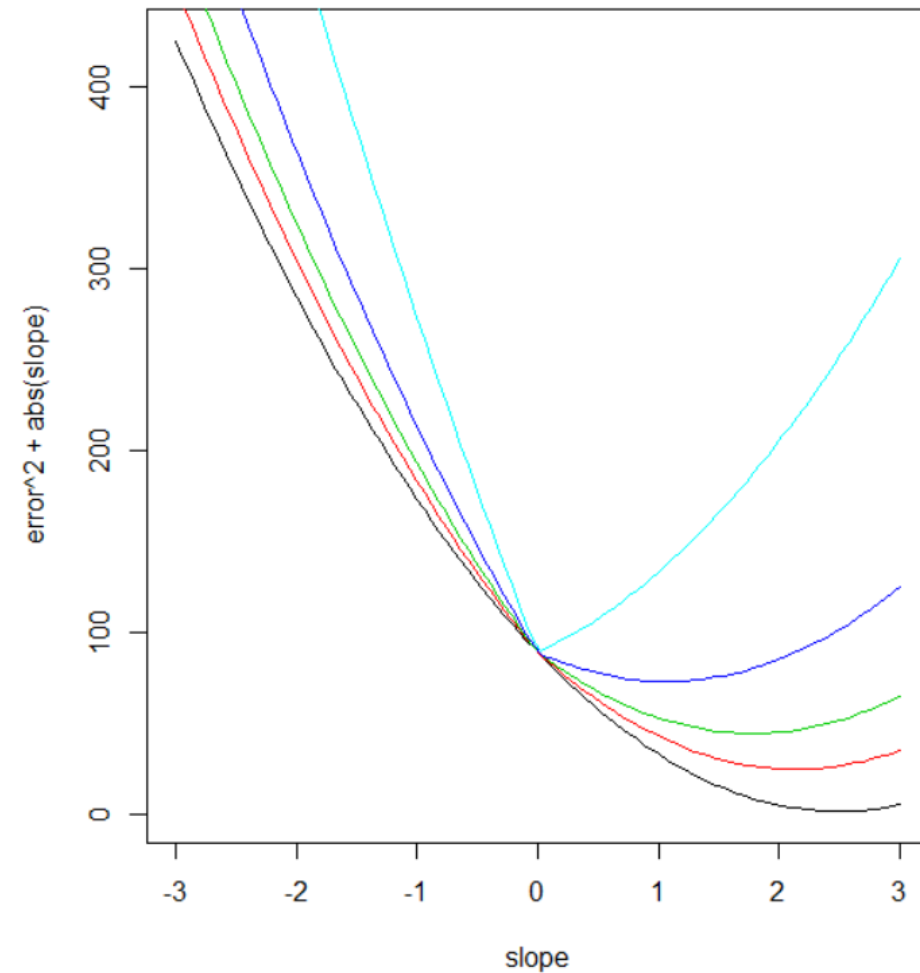
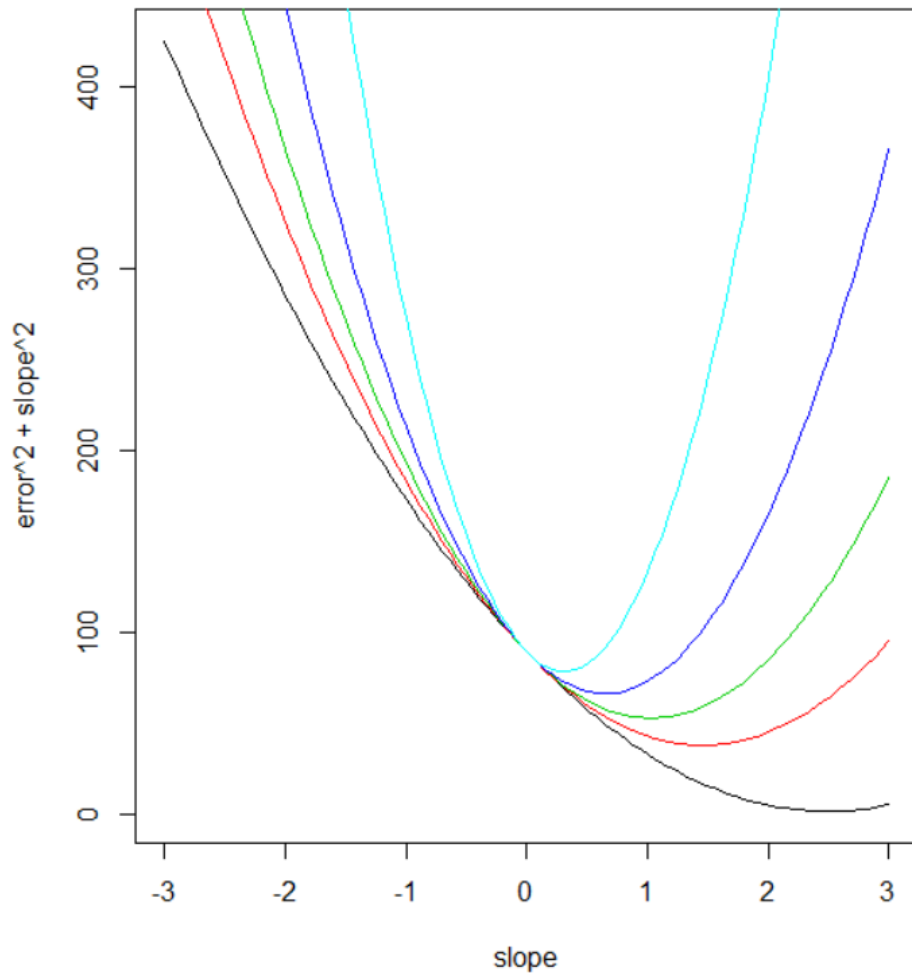
Regulation term(s) can be considered

$$\min_w \frac{1}{2} \|w\|^2 - C \cdot l(w)$$



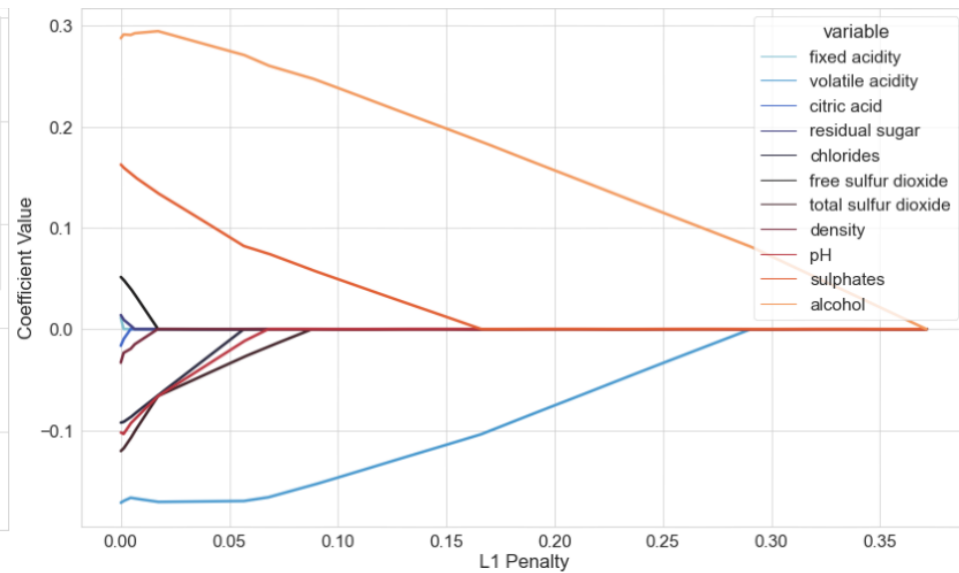
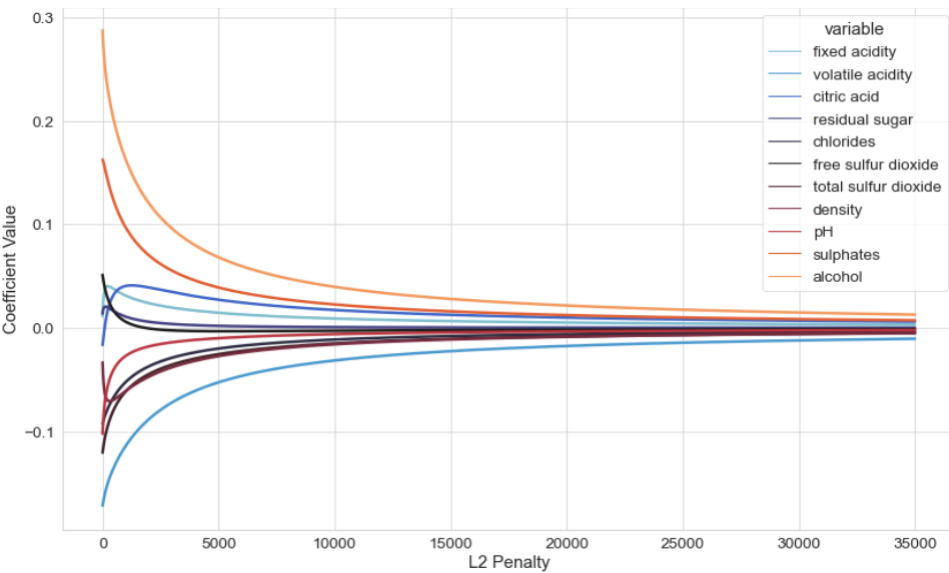


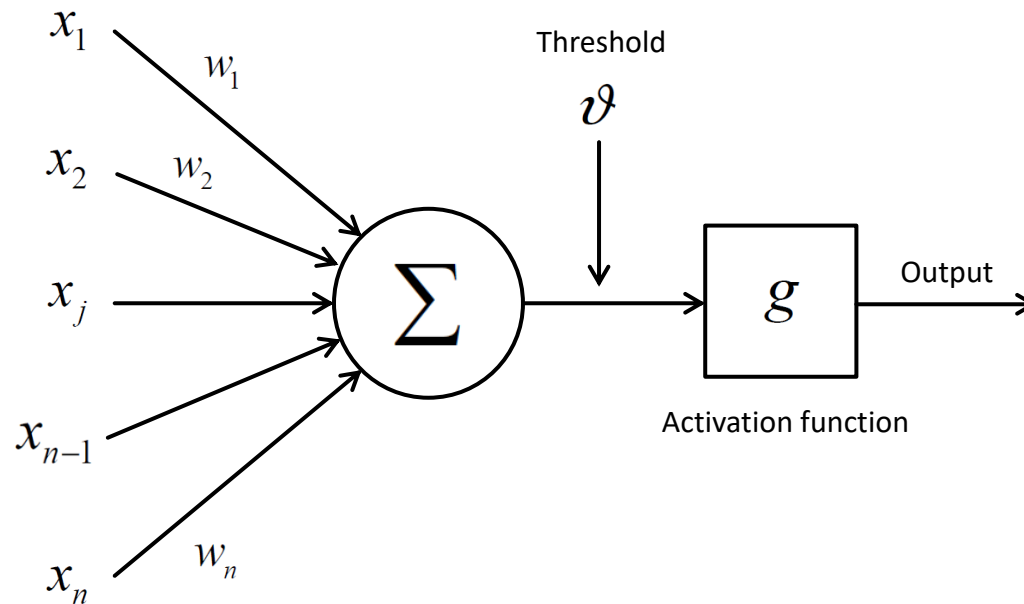
## Regularization



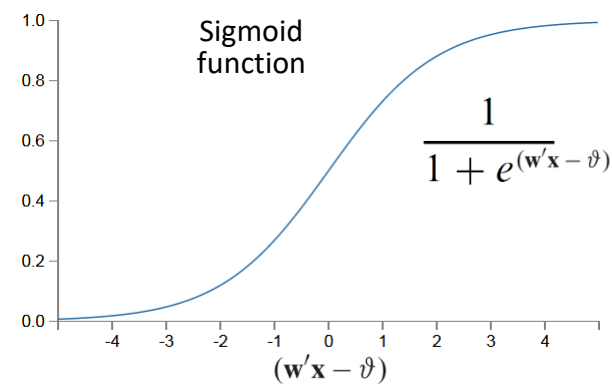
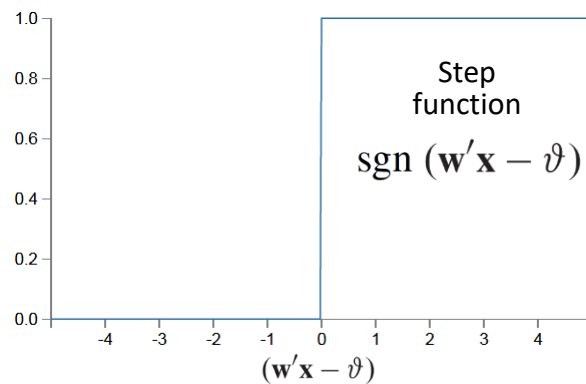


## Regularization



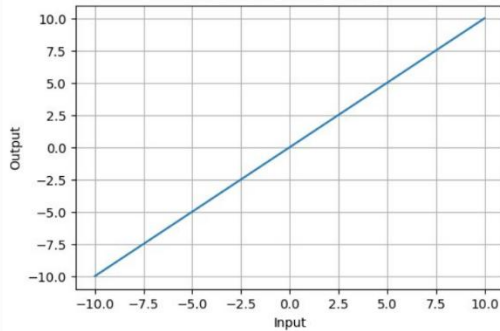


- 1 Initialize weights and threshold
- 2 Input an example and compute its output
- 3 Update weights and threshold according to the entity of the error made
$$\mathbf{W} = \mathbf{W} + \text{error } \mathbf{X}$$
$$\vartheta = \vartheta + \text{error}$$
- 4 Repeat steps 2 and 3 until the error at a given iteration is less than a threshold or a number of iterations have been completed



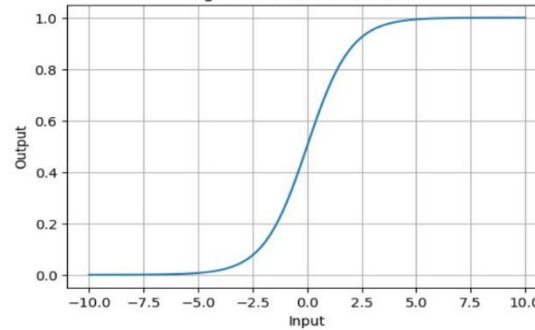


Linear Activation Function



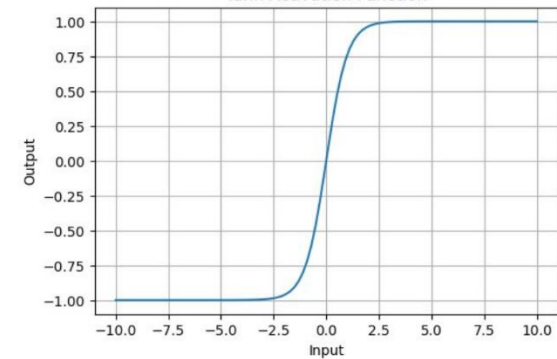
$$h(x) = x$$

Sigmoid Activation Function



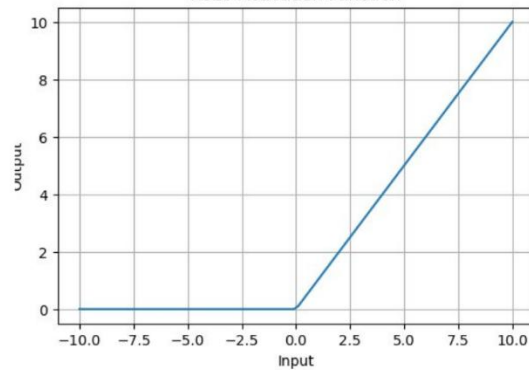
$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Tanh Activation Function



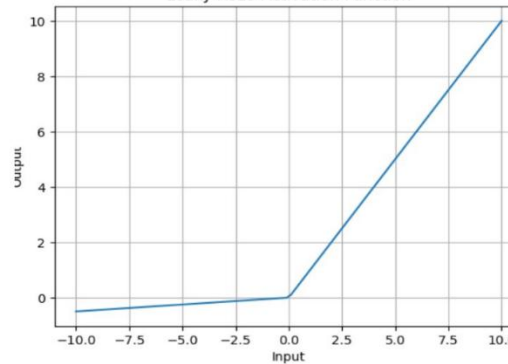
$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

ReLU Activation Function



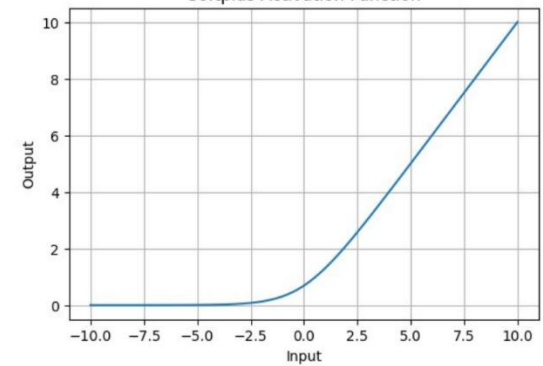
$$\text{ReLU}(x) = \max(0, x)$$

Leaky ReLU Activation Function



$$\text{Leaky ReLU}(x) = \begin{cases} 0.01x & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$$

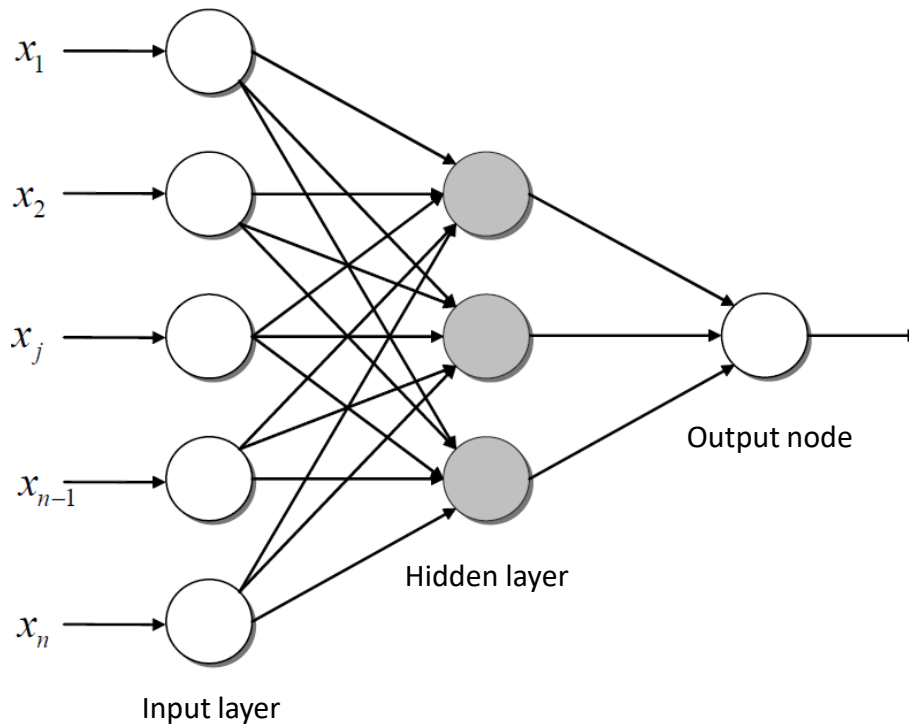
Softplus Activation Function



$$f(x) = \ln(1 + e^x)$$



Each hidden unit can be considered as a single perceptron.



- 1 Initialize weights and threshold.
- 2 Input an example, compute its output and the corresponding error.
- 3 Starting with the output layer propagate the error back to the previous layer and update the weights.
- 4 Repeat steps 2 and 3 for all the training examples.



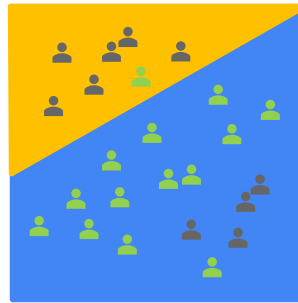
The perceptron generates classification functions in the form of separating hyperplanes.

What happens when the number of hidden layers/nodes increases?

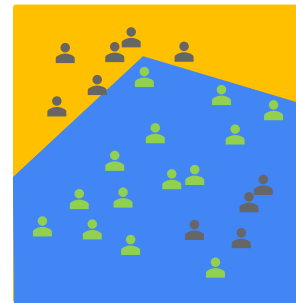
More complex classification regions are obtained!

You can play with this here:  
<https://playground.tensorflow.org>

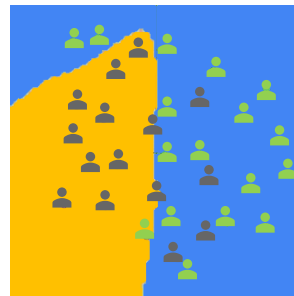
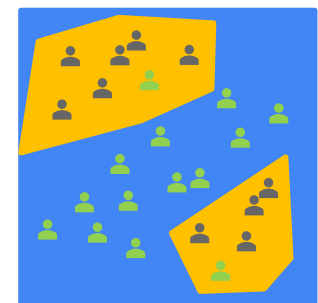
1 LAYER



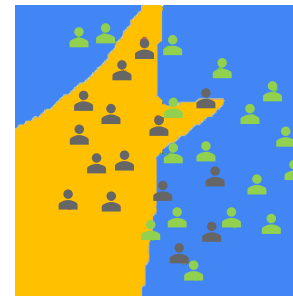
2 LAYERS



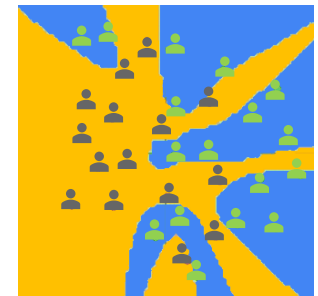
3+ LAYERS



1 LAYER, 3 NODES



1 LAYER, 6 NODES



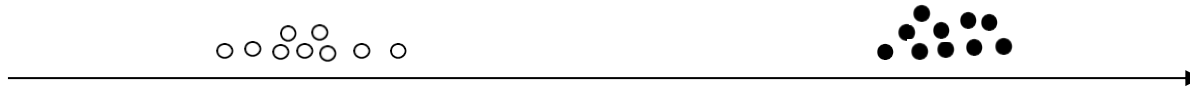
1 LAYER, 20 NODES

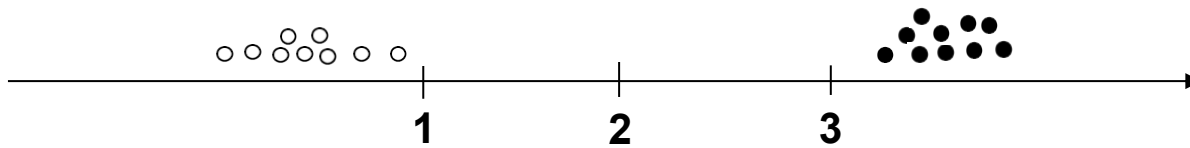


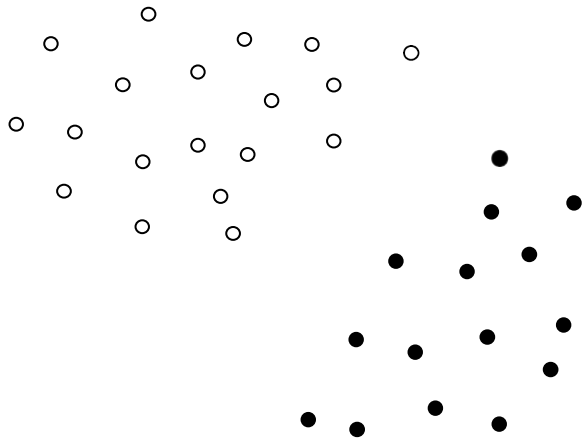
Some comments:

- ① Very effective techniques (deep learning)
- ② Training a network is an art...and can be extremely expensive
- ③ Interpretability is among the most prominent issues



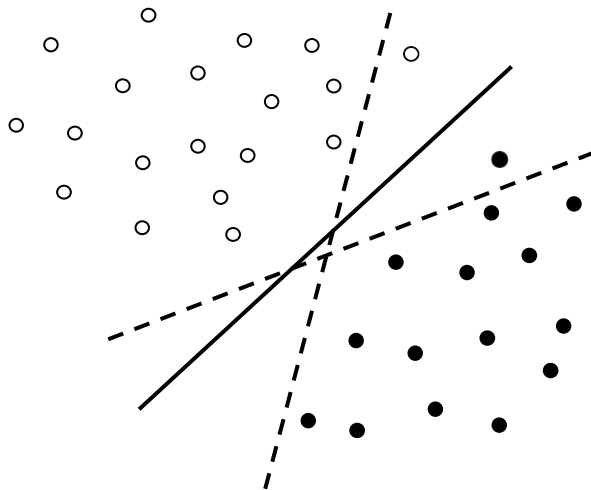






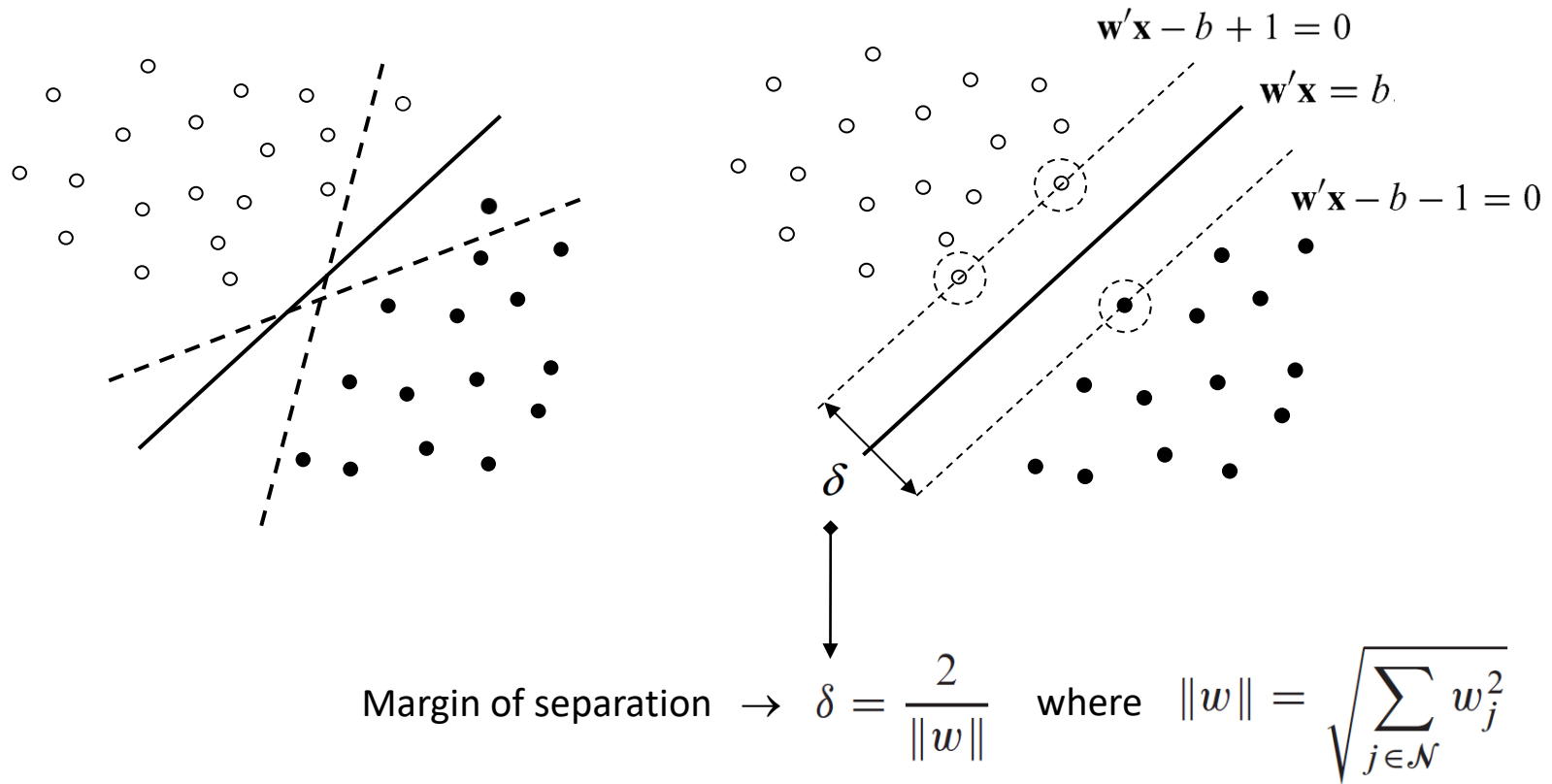


If the examples are linearly separable there exist infinite optimal separating hyperplanes...





If the examples are linearly separable there exist infinite optimal separating hyperplanes...

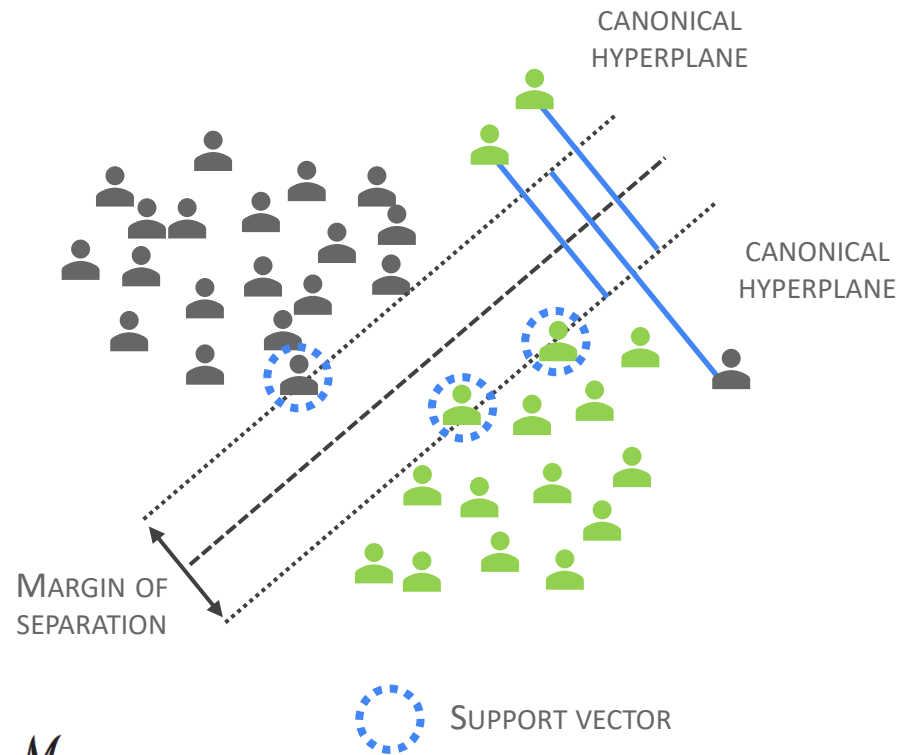


If the dataset is linearly separable, the optimal separating hyperplane is determined by solving the following optimization problem:

$$\begin{aligned} \min_{\mathbf{w}, b} \quad & \frac{1}{2} \|\mathbf{w}\|^2 \\ \text{s. a} \quad & y_i(\mathbf{w}\mathbf{x}_i - b) \geq 1 \quad i \in \mathcal{M} \end{aligned}$$

Allowing for soft margins:

$$\begin{aligned} \min_{\mathbf{w}, b, d} \quad & \frac{1}{2} \|\mathbf{w}\|^2 + \lambda \sum_{i=1}^m d_i \\ \text{s. a} \quad & y_i(\mathbf{w}\mathbf{x}_i - b) \geq 1 - d_i \quad i \in \mathcal{M} \\ & d_i \geq 0 \quad i \in \mathcal{M} \end{aligned}$$





How do we solve this quadratic problem? Lagrangian multipliers

$$L(\mathbf{w}, \alpha, b) = \frac{1}{2} \|\mathbf{w}\|^2 - \sum_{i \in N} \alpha_i [y_i (\mathbf{w} \cdot \mathbf{x}_i + b) - 1]$$

$$\frac{\partial L}{\partial \mathbf{w}} = \mathbf{w} - \sum_{i \in N} \alpha_i y_i \mathbf{x}_i = 0$$

$\Rightarrow \mathbf{w} = \sum_{i \in N} \alpha_i y_i \mathbf{x}_i$  i.e., weight are a linear sum of the samples

$$\frac{\partial L}{\partial b} = - \sum_{i \in N} \alpha_i y_i = 0$$

$$\Rightarrow \sum_{i \in N} \alpha_i y_i = 0$$

$$L(\mathbf{w}, \alpha, b) = \sum_{i \in N} \alpha_i - \frac{1}{2} \sum_{i \in N} \sum_{j \in N} \alpha_i \alpha_j y_i y_j \mathbf{x}_i' \mathbf{x}_j$$



What about nonlinear separations?

$$L(\mathbf{w}, \alpha, b) = \sum_{i \in N} \alpha_i - \frac{1}{2} \sum_{i \in N} \sum_{j \in N} \alpha_i \alpha_j y_i y_j \mathbf{x}_i' \mathbf{x}_j$$

$$L(\mathbf{w}, \alpha, b) = \sum_{i \in N} \alpha_i - \frac{1}{2} \sum_{i \in N} \sum_{j \in N} \alpha_i \alpha_j y_i y_j \phi(\mathbf{x}_i)' \phi(\mathbf{x}_j)$$





What about nonlinear separations?

- Let's consider a second degree polynomial mapping:

$$\phi(\mathbf{x}) \stackrel{\text{e.g.}}{=} \phi\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) \stackrel{\text{e.g.}}{=} \begin{pmatrix} x_1^2 \\ \sqrt{2}x_1x_2 \\ x_2^2 \end{pmatrix}$$

- This, however, can lead to operations with vectors of very high dimensionality.
- But we are only interested in the result of  $\phi(\mathbf{x}) \cdot \phi(\mathbf{z}) \rightarrow$  Kernel trick
- Kernel function:

$$K(\mathbf{x}, \mathbf{z}) = \phi(\mathbf{x}) \cdot \phi(\mathbf{z})$$



For a second degree polynomial mapping:

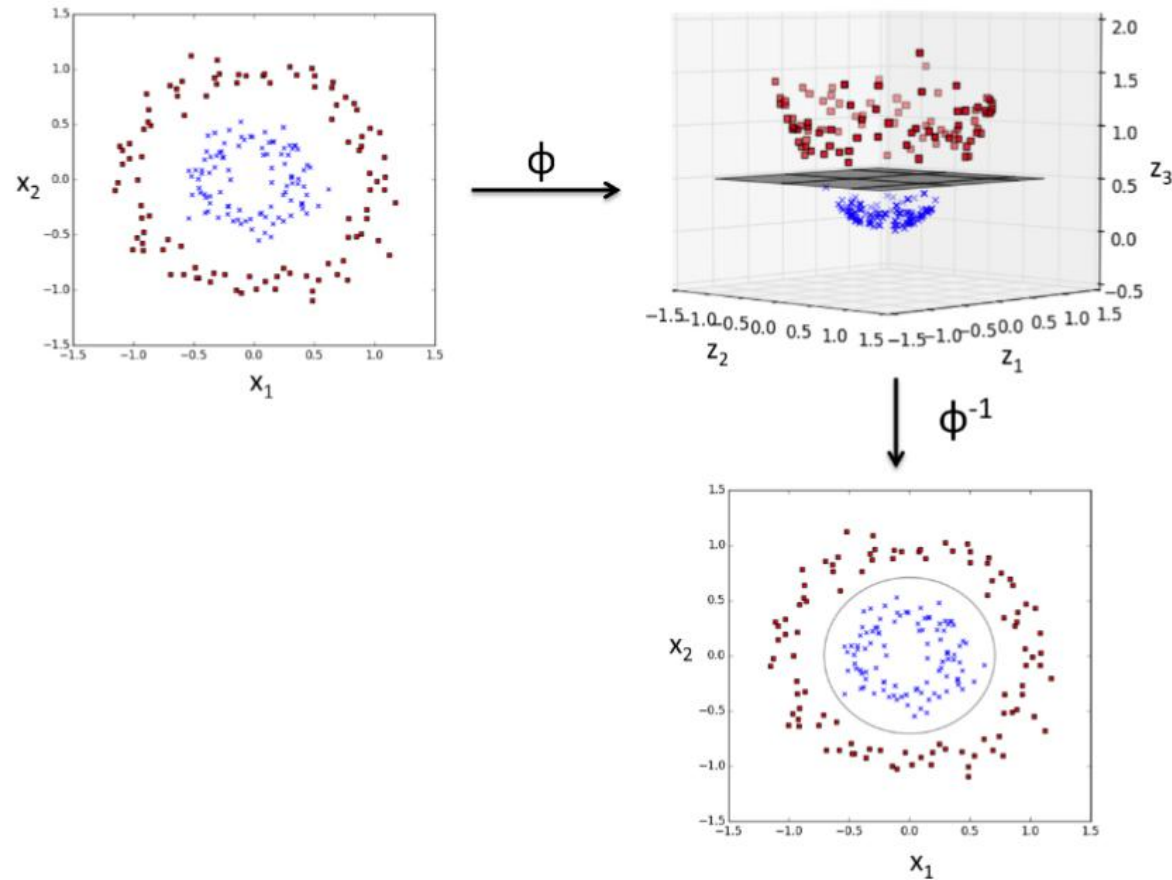
$$\begin{aligned}\phi(\mathbf{x}) \cdot \phi(\mathbf{z}) &= \begin{pmatrix} x_1^2 \\ \sqrt{2}x_1x_2 \\ x_2^2 \end{pmatrix}^T \begin{pmatrix} z_1^2 \\ \sqrt{2}z_1z_2 \\ z_2^2 \end{pmatrix} \\ &= x_1^2z_1^2 + 2x_1z_1x_2z_2 + x_2^2z_2^2 \\ &= (x_1z_1 + x_2z_2)^2 = \left( \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}^T \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \right)^2 = (\mathbf{x} \cdot \mathbf{z})^2\end{aligned}$$

Some well known kernels:

- linear:  $x'x$
- poly:  $(\gamma x'x + r)^d$
- rbf:  $e^{(-\gamma \|x-x'\|^2)}$
- sigmoid:  $\tanh(\gamma x'x + r)$

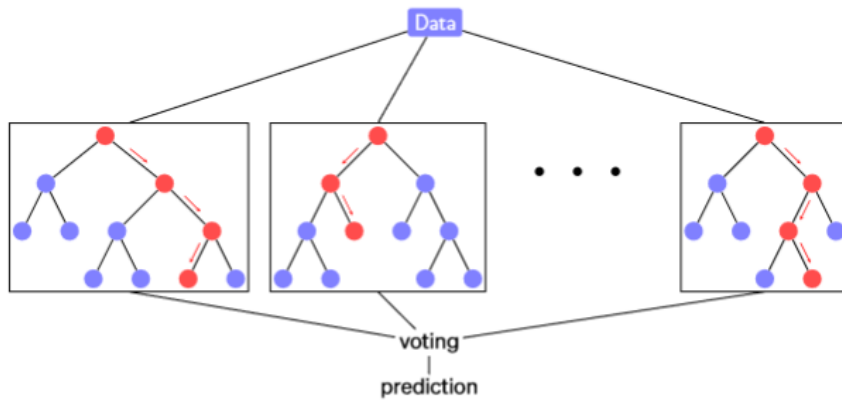


What about nonlinear separations?

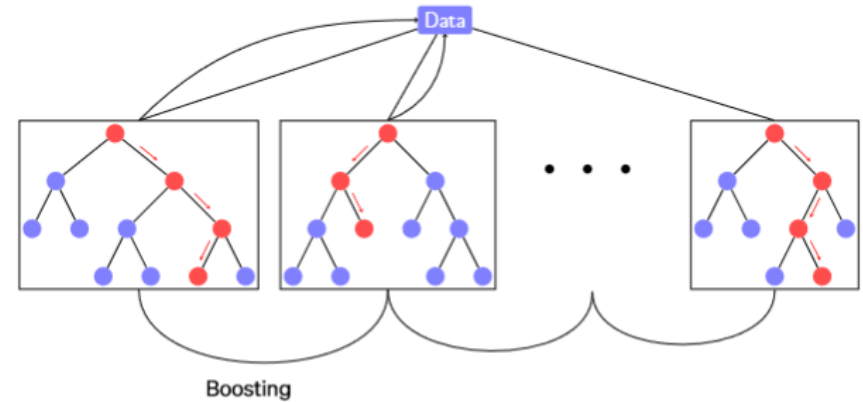




## Bagging

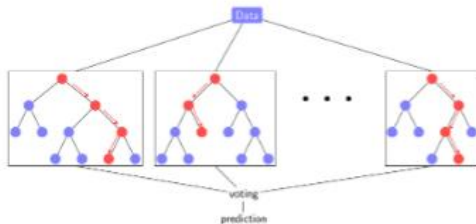


## Boosting





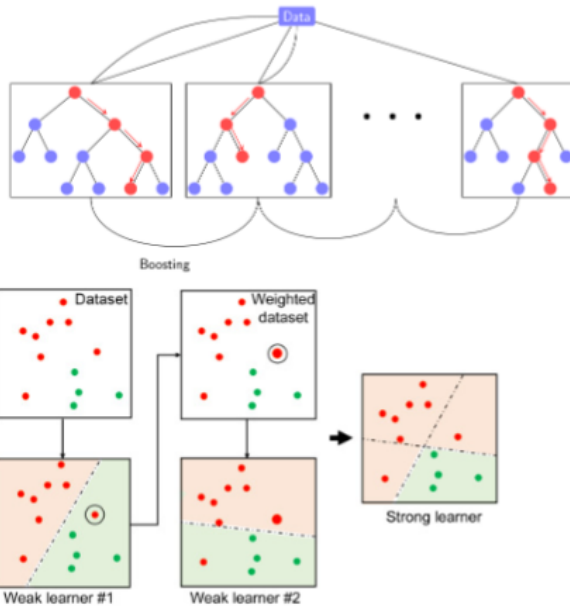
**Bagging (bootstrap aggregating)** is an ensemble framework aimed at improving the stability and accuracy of a classifier. It helps to avoid overfitting.



1. Create different (simple) tree models (stumps)
2. Each model is created with a subset of observation/features ( $\sim 2m/3$ )
3. We combine the prediction of all trees

## Main Parameters

- ▶ `n_estimators`: Number of trees
- ▶ `max_features`: Number of features selected for the split
- ▶ `bootstrap=False`: Use all samples
- ▶ Tree parameters



1. Assign equal weights to observations  $w_i^{(0)} = 1/m$
2. For  $k = 1, \dots, K$ 
  - Select a sample of observations based on the weights.
  - Create the  $k$ -th weak learner and compute predictions  $x^{(k)}$
  - Compute the model **weighted** error and assign its coefficient:

$$\alpha^{(k)} = \lambda \times \log((1 - \text{error})/\text{error})$$

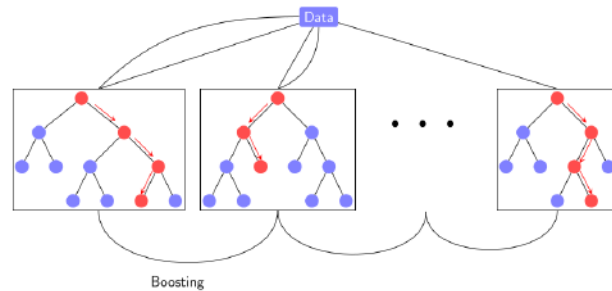
- Update sample weights:

$$w_i^{(k+1)} \propto w_i^{(k)} \times \exp(-\alpha^{(k)} y_i \hat{x}_i^{(k)})$$

3. Final weighted prediction

## Main Parameters

- `n_estimators`: Number of estimators ( $K$ )
- `base_estimator`: Weak estimator type
- `learning_rate`: weights of estimator in final decision ( $\lambda$ )



1. Train a weak learner  $F_0$  and compute predictions  $\hat{y}_0 = F_0(x)$
2. For  $k = 1, \dots, K$ 
  - Compute the difference between the target ( $y$ ) and the prediction of the current learner ( $\hat{y}_{k-1}$ )
  - Train a weak learner that minimizes the loss function (error):

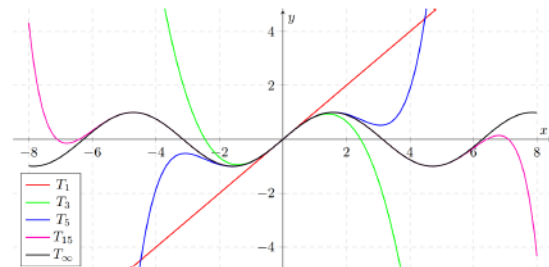
$$P_k(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \dots + \frac{f^{(k)}(a)}{k!}(x - a)^k$$

$$f_k = \arg \min_f L_m = \arg \min_f \sum_{i=1}^n l(y_i, F_{k-1}(x_1) + f(x_i))$$

- $F_k = F_{k-1} + \lambda f_k$

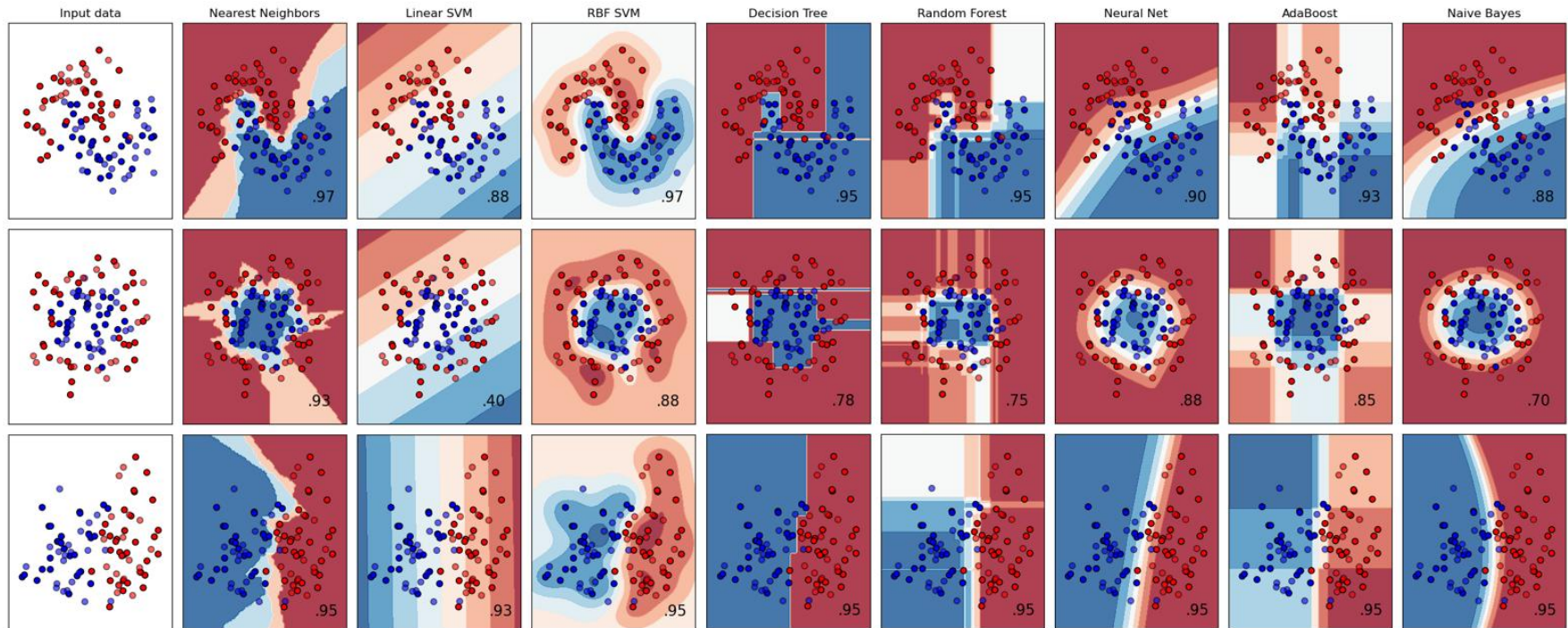
## Main Parameters

- ▶ n\_estimators: Number of estimators ( $K$ )
- ▶ base\_estimator: Weak estimator type
- ▶ learning\_rate: weights of estimator in final decision ( $\lambda$ )



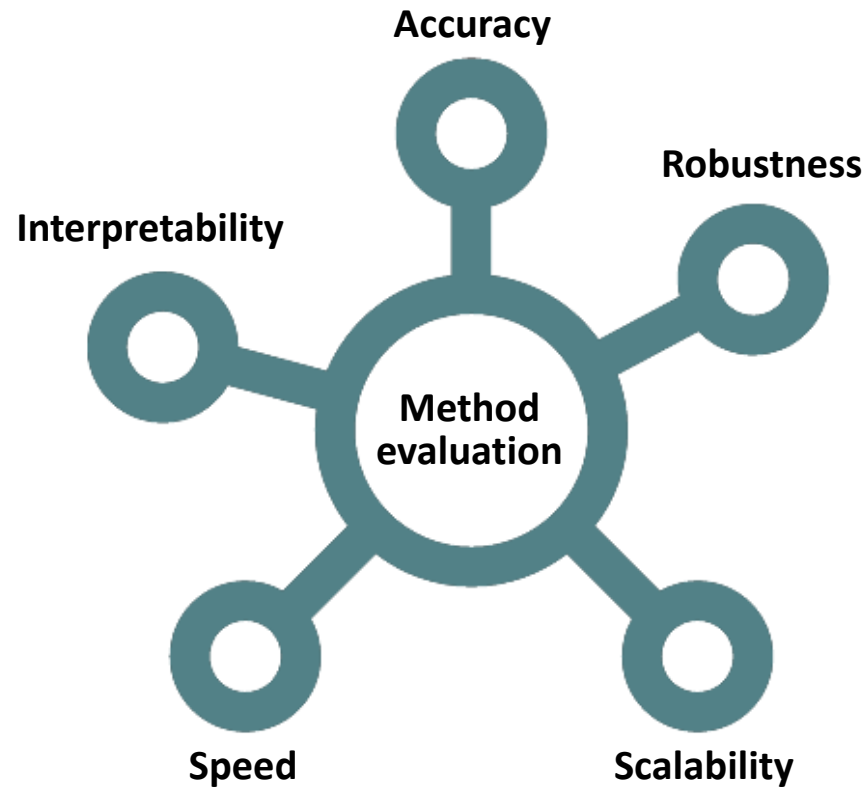


# How do classifiers behave?



Source: sklearn, classifier comparison



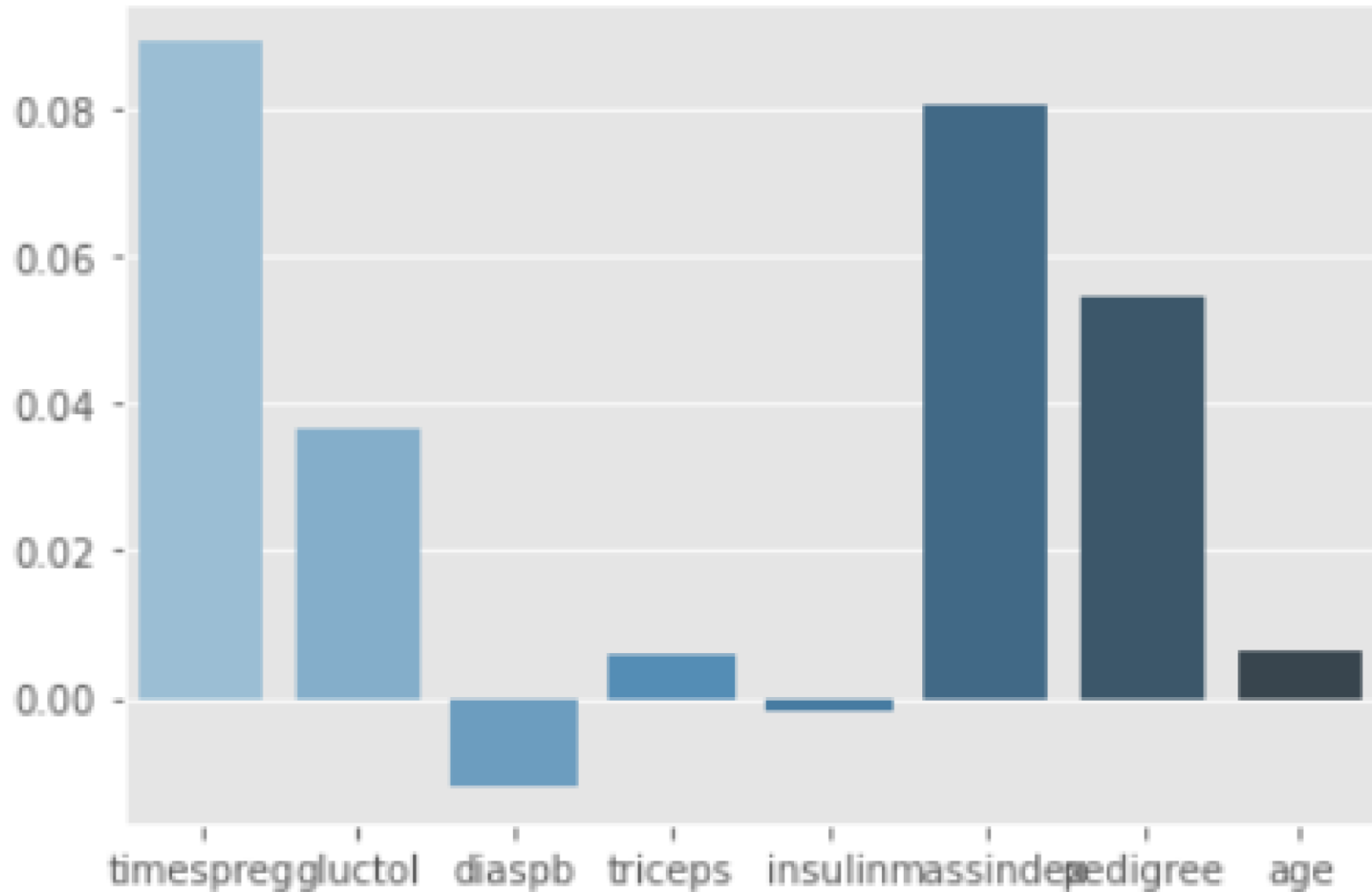




## Logistic Regression

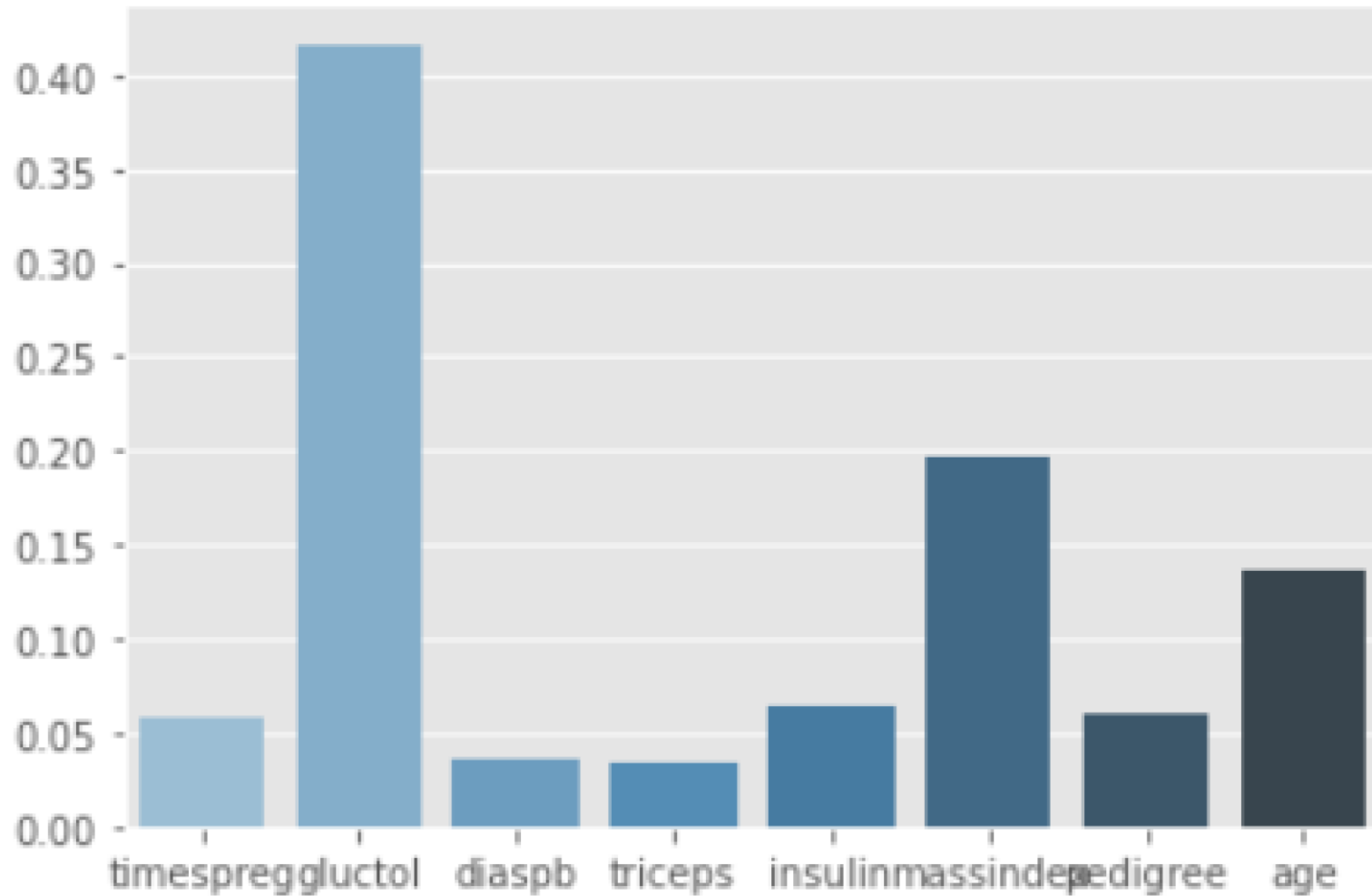


## Logistic Regression



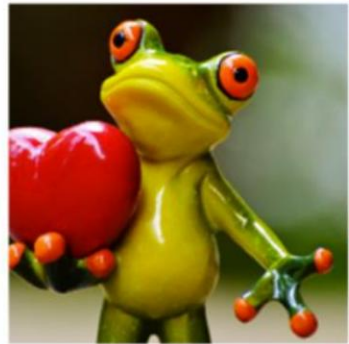


Random forests (through decision trees)





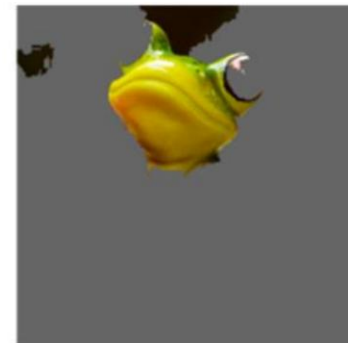
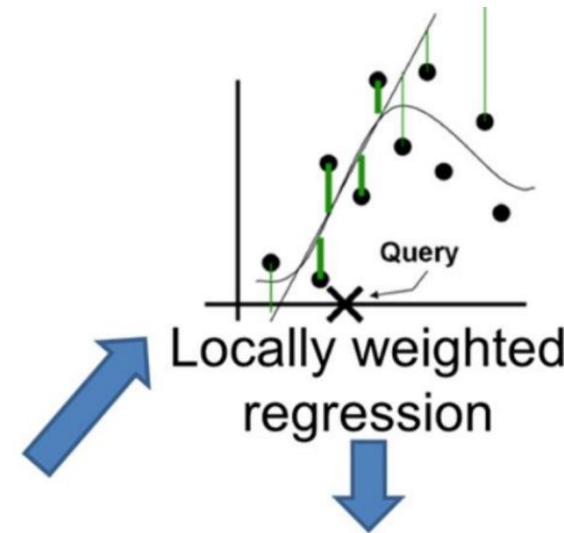
What if it is not natively supported? Option 1: **LIME**



Original Image

$P(\text{tree frog}) = 0.54$

| Perturbed Instances | $P(\text{tree frog})$ |
|---------------------|-----------------------|
|                     | <br>0.85              |
|                     | <br>0.00001           |
|                     | <br>0.52              |



Explanation



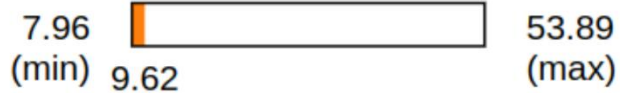
## Local Interpretable Model-agnostic Explanations

```
exp_mlp = explainer.explain_instance(X_train[i], mlp.predict, num_features=5)
```

```
Intercept 25.123228322789352  
Prediction_local [15.31616231]  
Right: 9.621792242553637
```

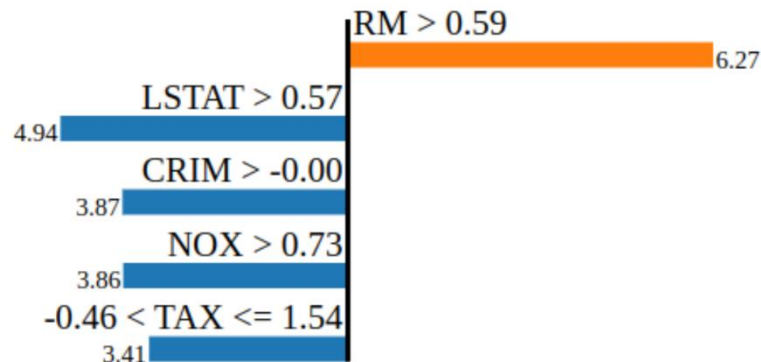
```
exp_mlp.show_in_notebook(show_table=True)
```

Predicted value



negative

positive



Feature Value

|       |      |
|-------|------|
| RM    | 0.82 |
| LSTAT | 1.82 |
| CRIM  | 0.83 |
| NOX   | 1.07 |
| TAX   | 1.54 |



## Option 2: Shapley Values

$$\phi_j(val) = \sum_{S \subseteq \{x_1, \dots, x_p\} \setminus \{x_j\}} \frac{|S|! (p - |S| - 1)!}{p!} (val(S \cup \{x_j\}) - val(S))$$



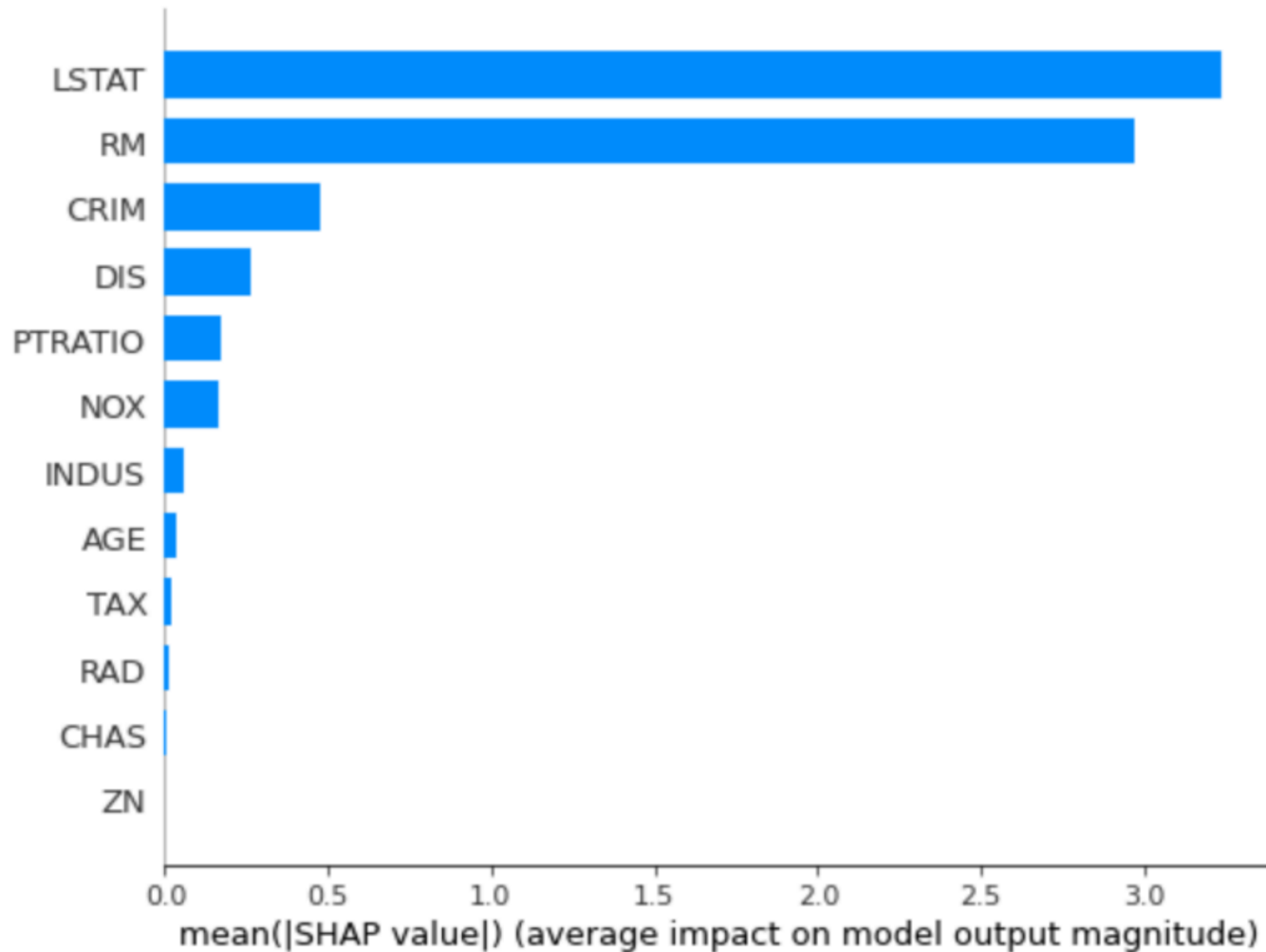
## Option 2: Shapley Values

$$\phi_j(val) = \sum_{S \subseteq \{x_1, \dots, x_p\} \setminus \{x_j\}} \frac{|S|! (p - |S| - 1)!}{p!} (val(S \cup \{x_j\}) - val(S))$$



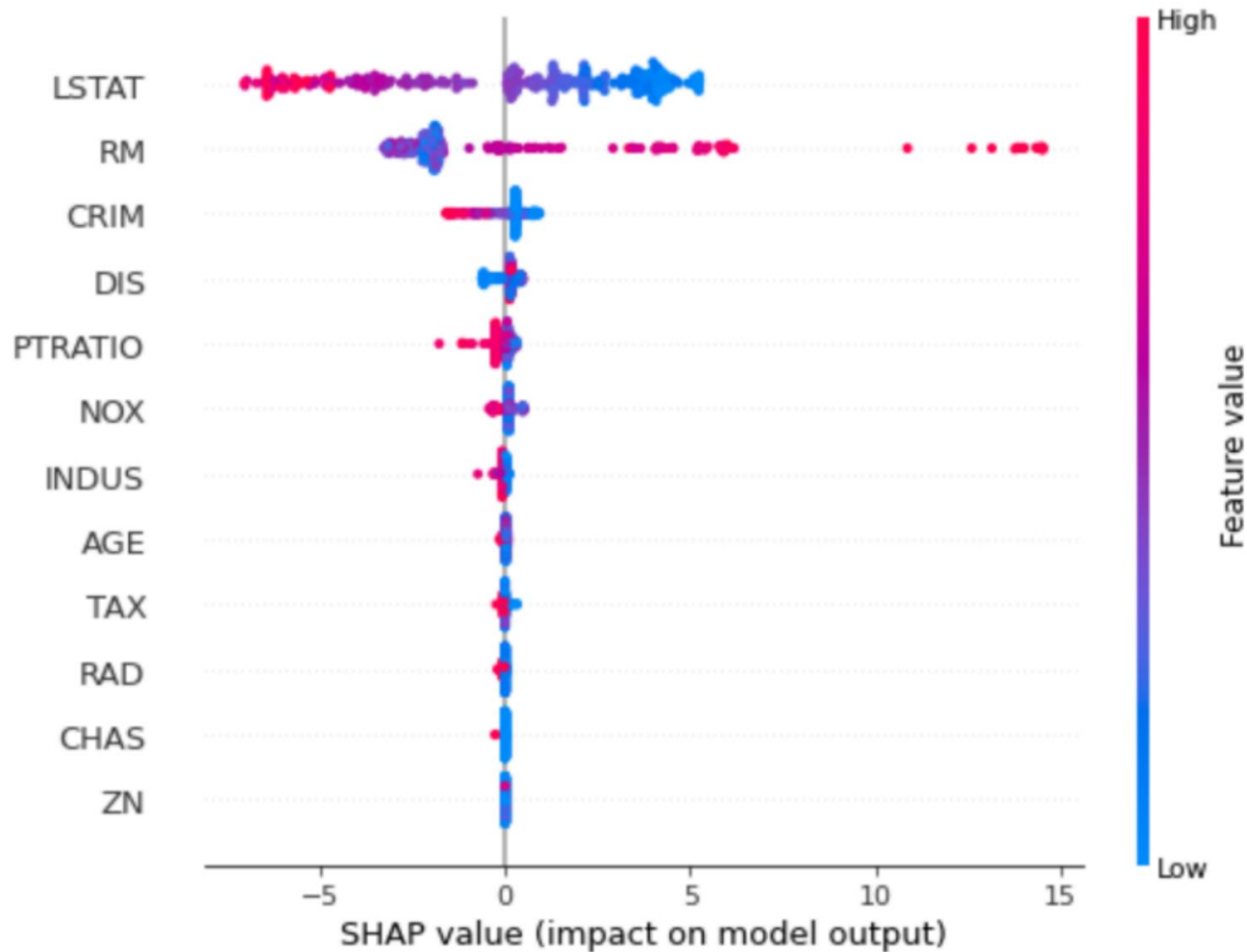


## Option 2: Shapley Values





## Shapley Values





## Shapley Values

```
shap.force_plot(explainer_rf.expected_value, shap_values, features=X_train, feature_names=df_X_train.columns)
```



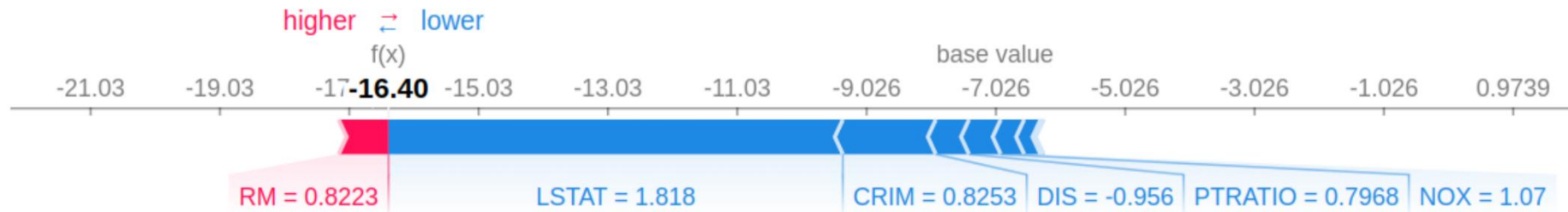


## Shapley Values

`i=0`

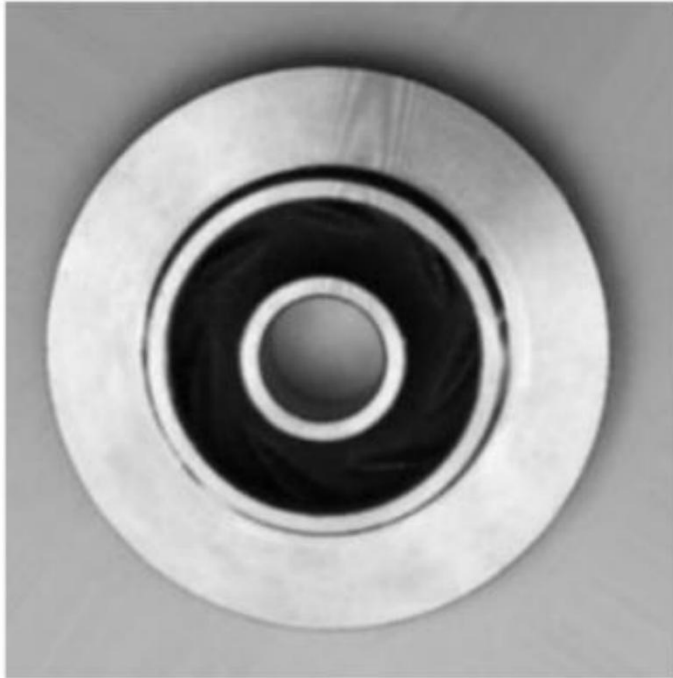
```
expected_value = shap_values[i, -1]
```

```
shap.force_plot(expected_value, shap_values[i, :], df_X_train.iloc[i, :])
```





**ok**



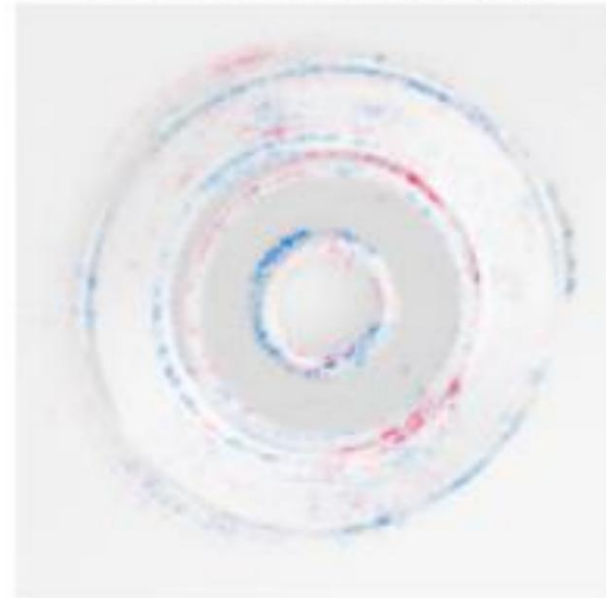
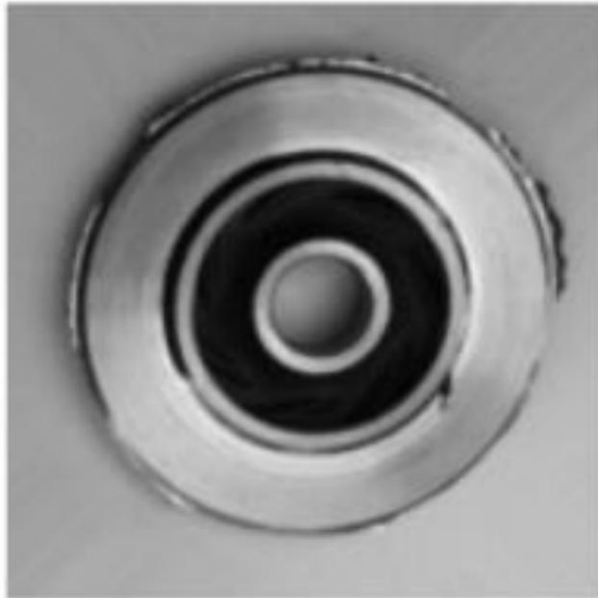
**def**

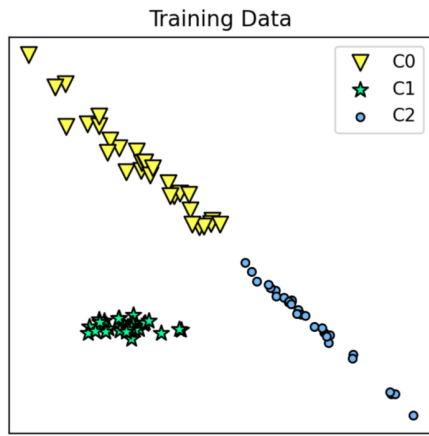


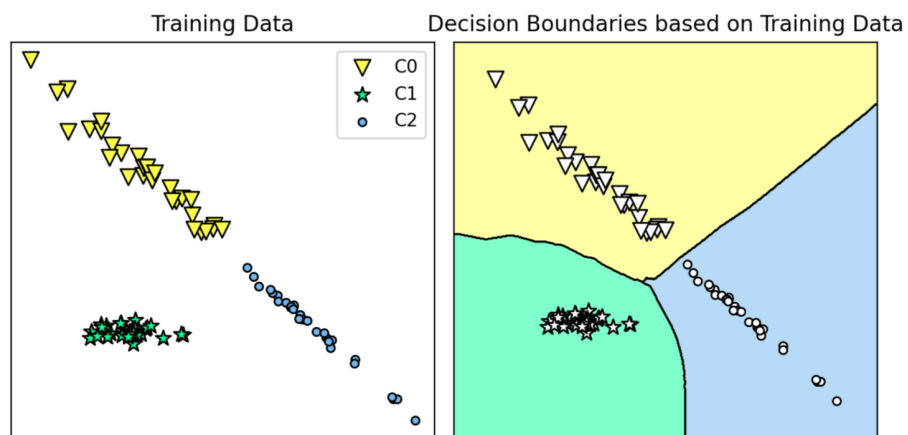


# Applications and (overcoming) limitations

cast\_def\_0\_1238.jpeg  
Actual Label : def



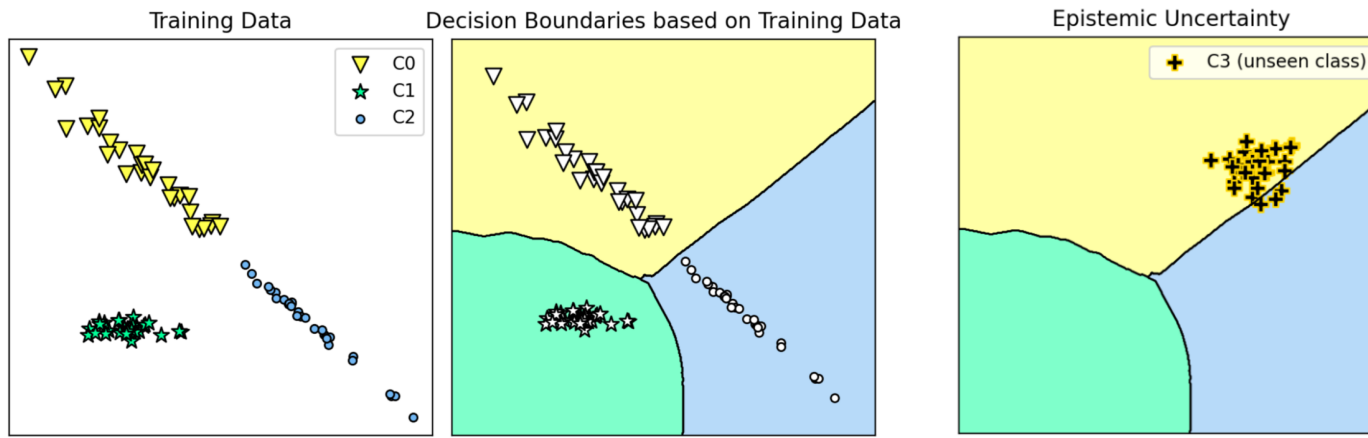


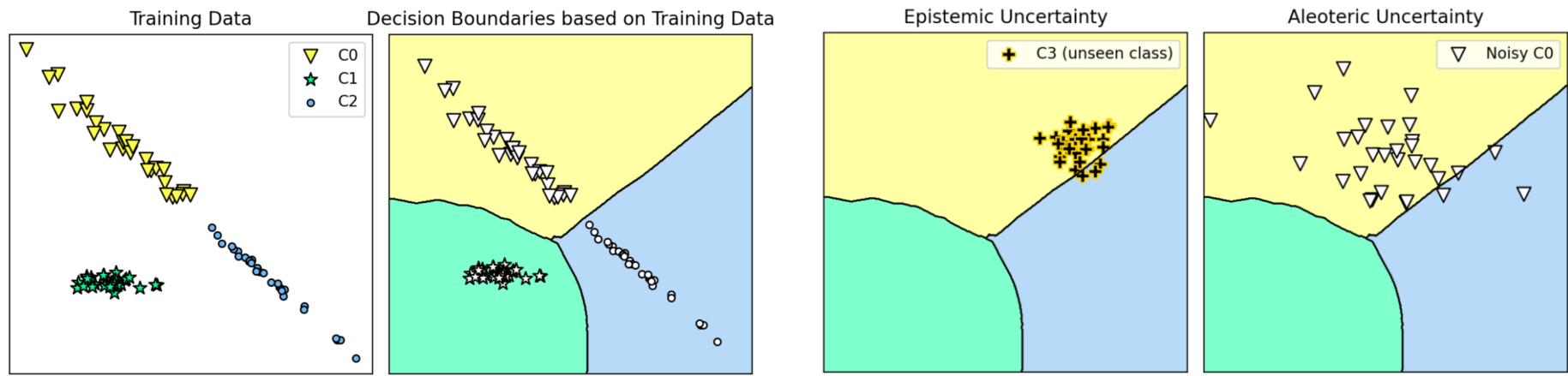






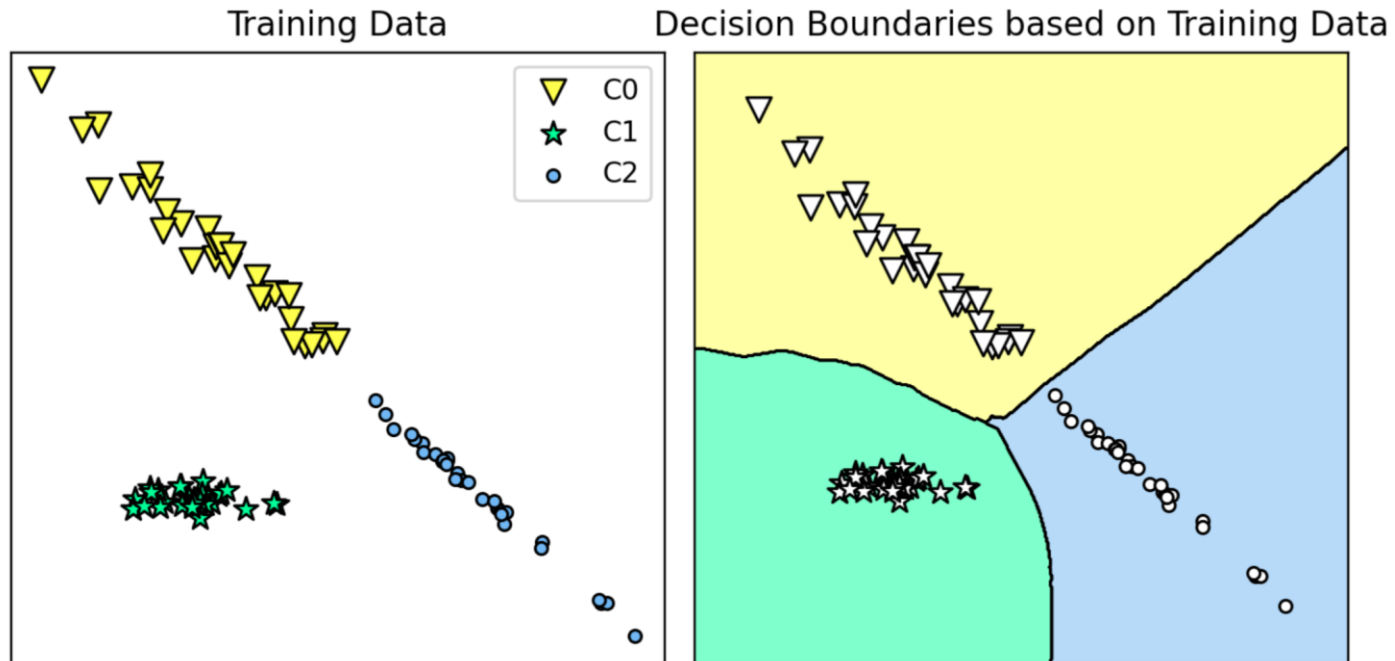
# Applications and (overcoming) limitations





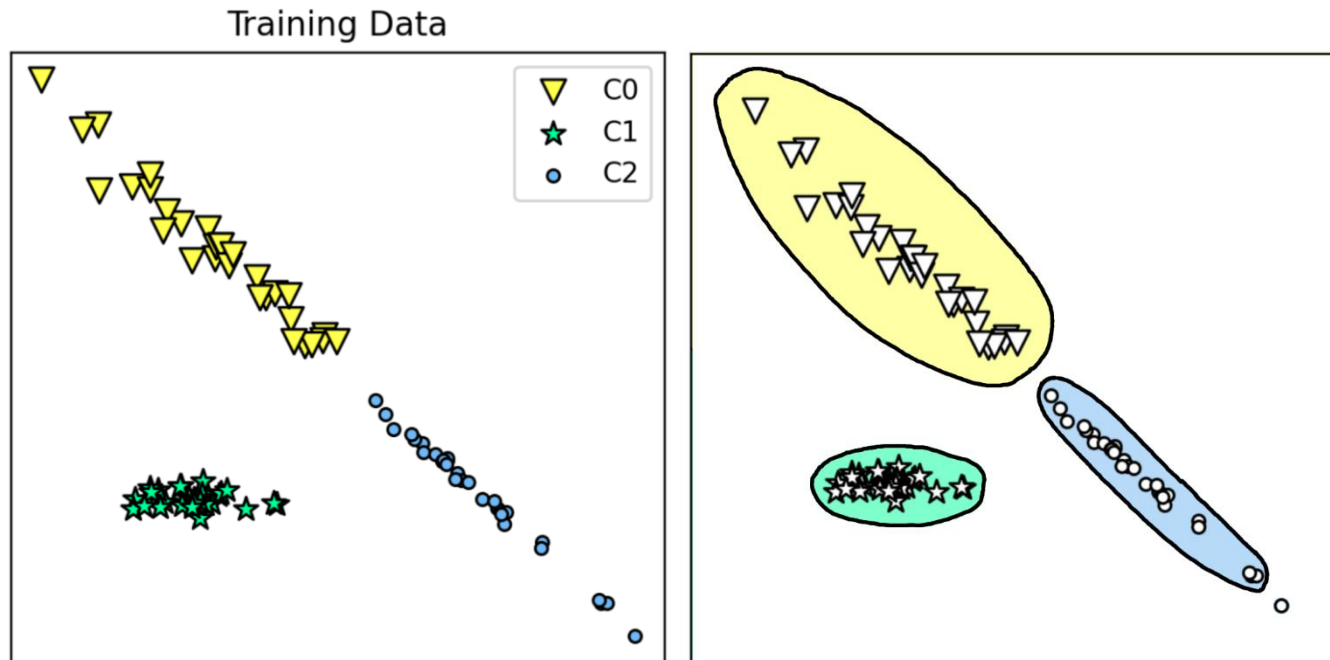


## Uncertainty-aware classification



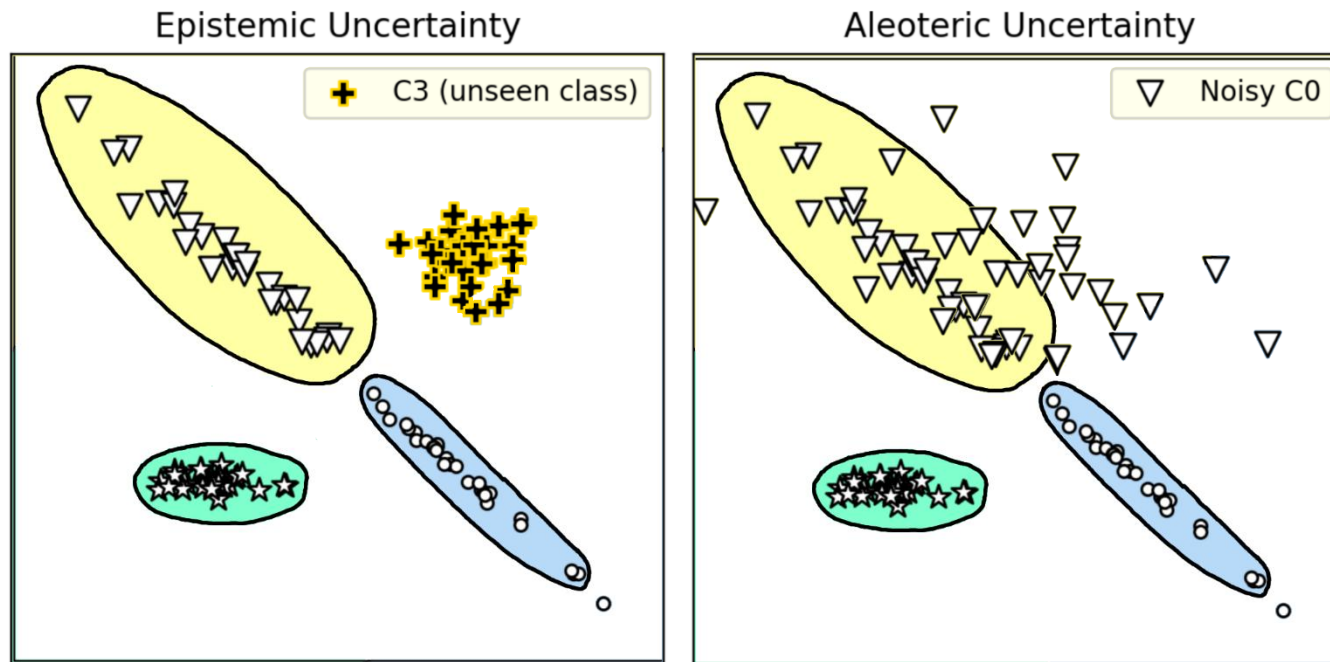


## Uncertainty-aware classification



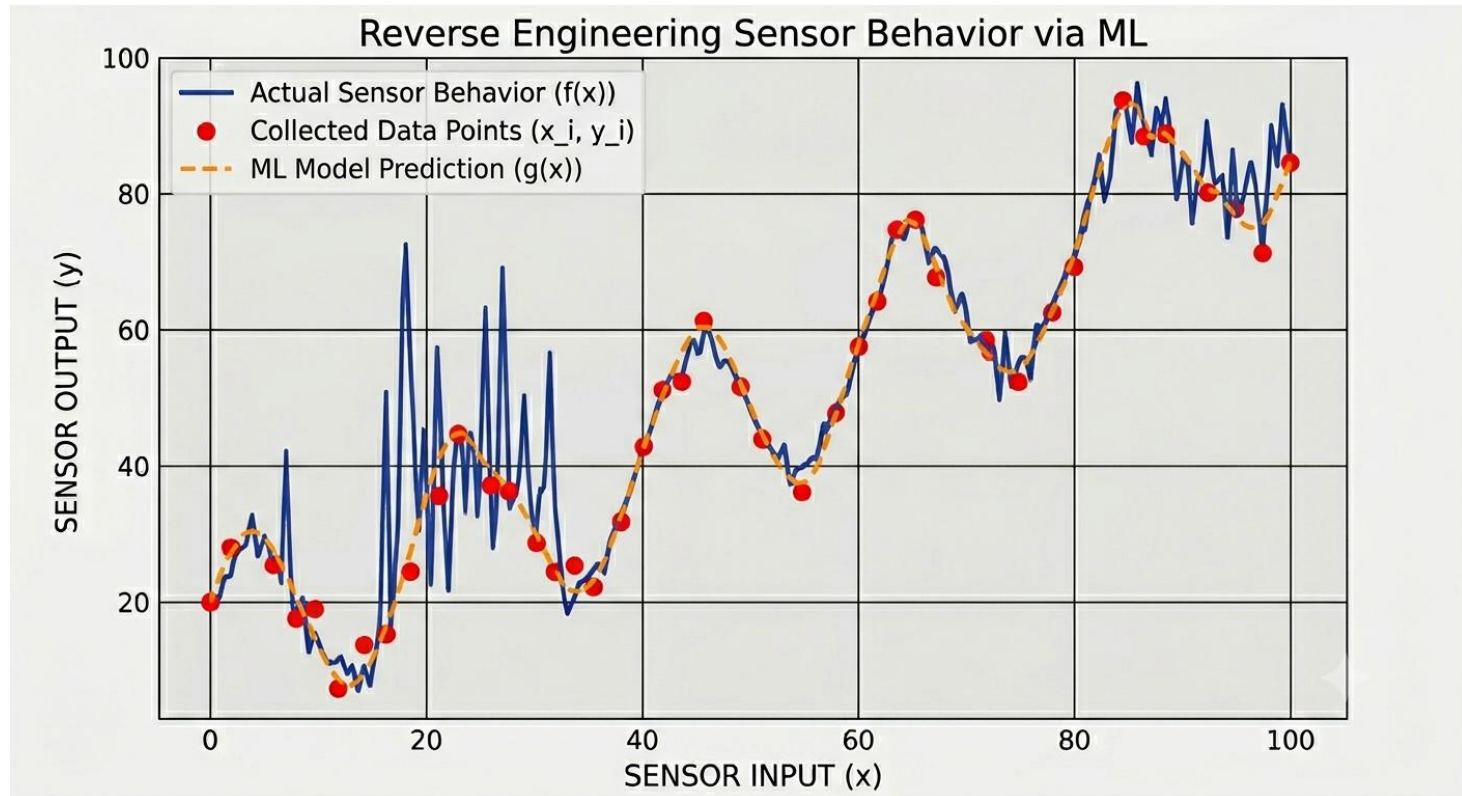


## Uncertainty-aware classification





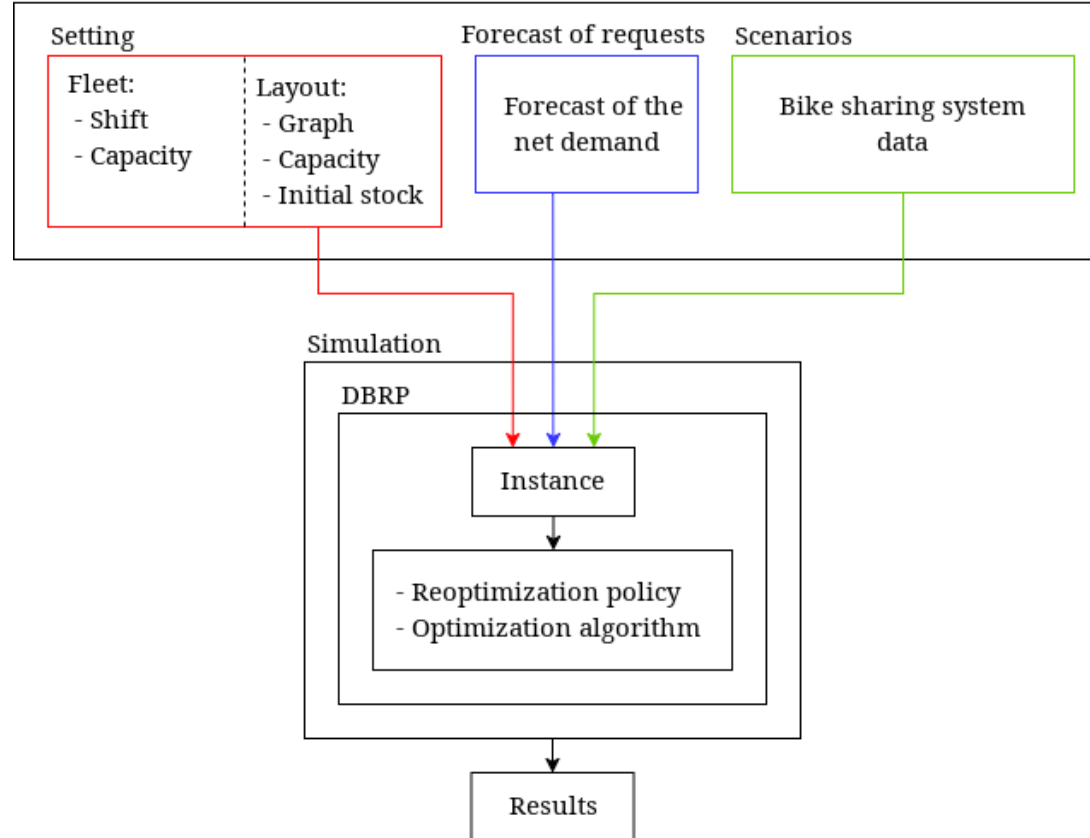
## Component simulation and reverse engineering





## Prediction and prescription

Input of the simulation framework





## Prediction and prescription

|       | Cost | Pred. 1 | Pred. 2 |
|-------|------|---------|---------|
| $c_1$ | 4    | 5       | 2       |
| $c_2$ | 5    | 4       | 7       |



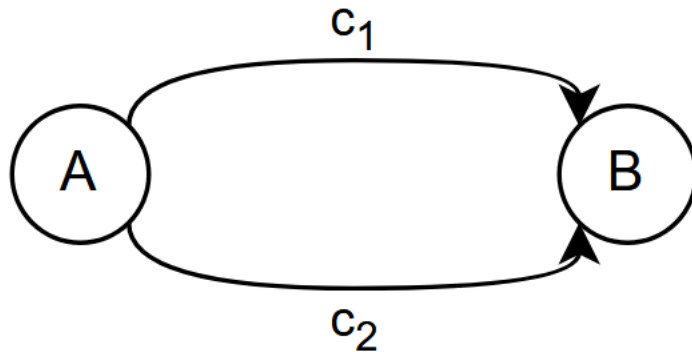


## Prediction and prescription

|       | Cost | Pred. 1 | Pred. 2 |
|-------|------|---------|---------|
| $c_1$ | 4    | 5       | 2       |
| $c_2$ | 5    | 4       | 7       |
| MSE   | -    | 1       | 4       |



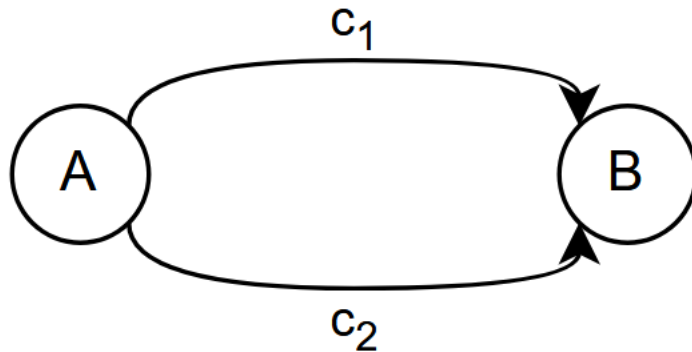
## Prediction and prescription



|       | Cost | Pred. 1 | Pred. 2 |
|-------|------|---------|---------|
| $c_1$ | 4    | 5       | 2       |
| $c_2$ | 5    | 4       | 7       |
| MSE   | -    | 1       | 4       |



## Prediction and prescription



|          | Cost | Pred. 1 | Pred. 2 |
|----------|------|---------|---------|
| $c_1$    | 4    | 5       | 2       |
| $c_2$    | 5    | 4       | 7       |
| MSE      | -    | 1       | 4       |
| Opt. gap | -    | 1       | 0       |



## Prediction and prescription

